

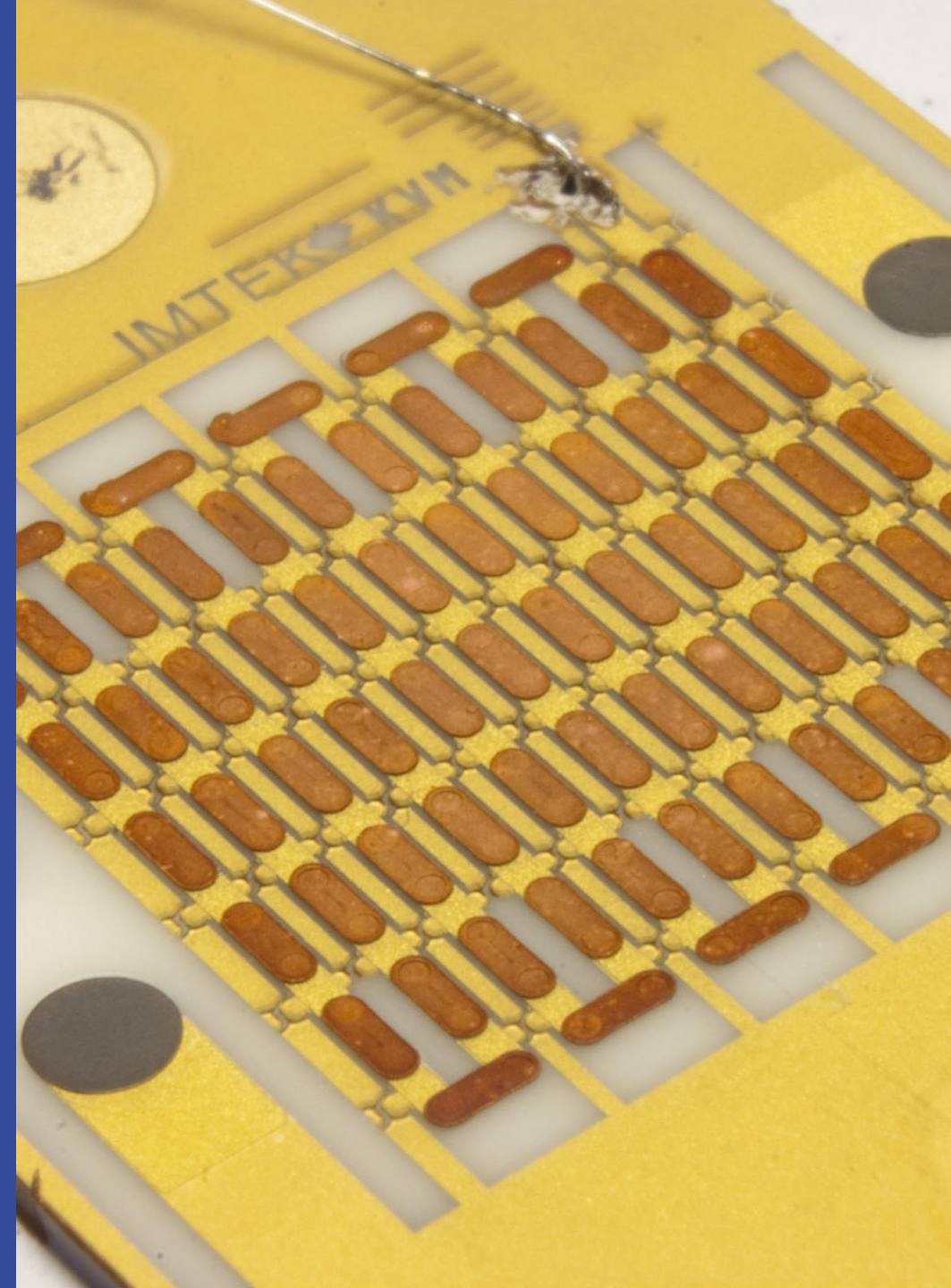
Micro Energy Harvesting

Chapter 2: Solar Energy Harvesting

Micro Energy Harvesting SoSe 2026

IMTEK – Lehrstuhl für Konstruktion von Mikrosystemen

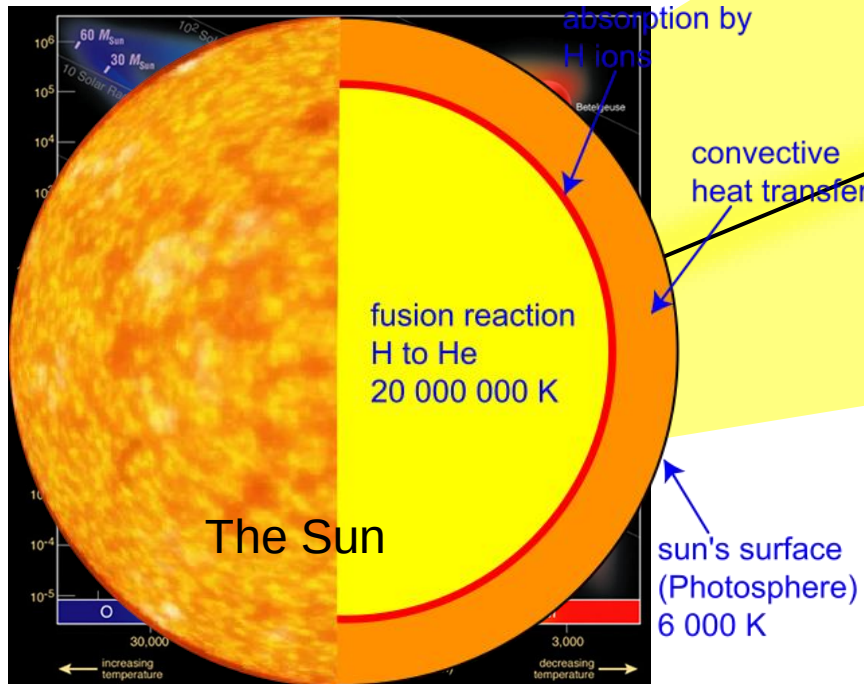
Prof. Dr. Peter Woias, Dr. Suman Kundu, Dr. Frank Goldschmidtboïng



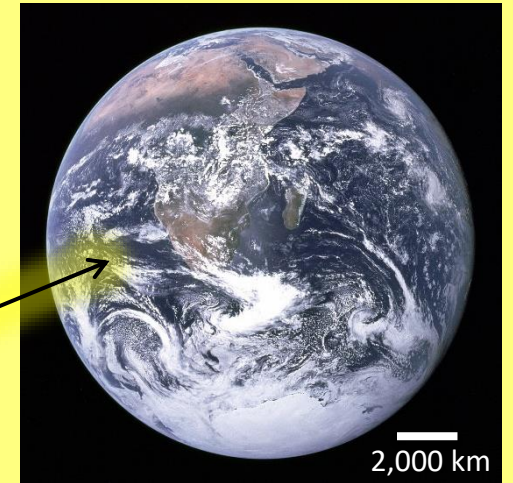
Solar Radiation

Basic Data

- luminosity: $3,86 \cdot 10^{26}$ W (power density: 140 W/m^3)
- age of sun ~ 4.57 billion years
- expected death of sun in ~ 7 billion years



149.6 Mio. km
 1367 W/m^2
 (= 1 solar constant)
 (@ earth AM0,
 UV, VIS, IR)



The Blue Marble, Apollo 17

$$H_0 = \frac{R_{sun}^2}{D^2} H_{sun}$$

$\sim 10\%$ $\sim 40\%$ $\sim 50\%$ of total radiation energy

Planck's Radiation Law

$$F(\lambda) = \frac{2\pi hc^2}{\lambda^5 \left(\exp\left(\frac{hc}{k\lambda T}\right) - 1 \right)}$$

Blackbody Power Density

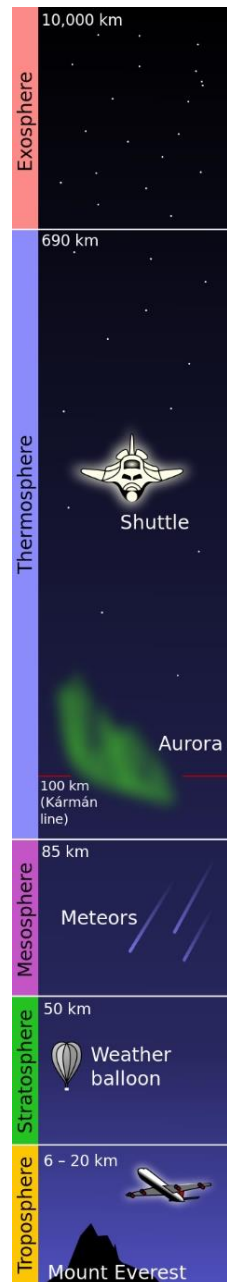
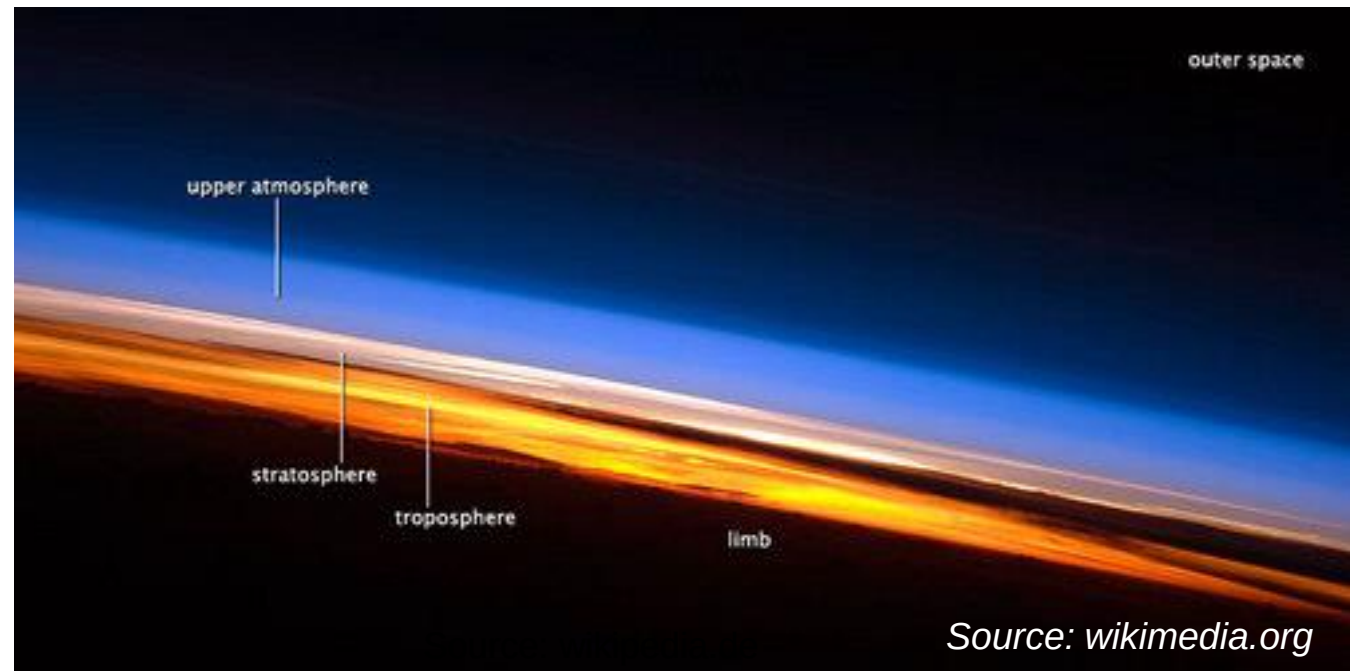
$$H = \sigma T^4$$

pveducation.org
 eso.org
 wikipedia.org

Solar Radiation

Air Mass Spectra (AM)

- incoming solar radiation (AM0) is absorbed and scattered by the atmosphere
- the **air mass (AM) in the path of the radiation** is a measure of how far it has to travel before reaching the ground
- the AM1 spectrum is the perfect vertical case
- the standard for tests of solar cells is the **AM1.5** spectrum
 - AM1.5D: direct radiation only
 - **AM1.5G** (global): additional radiation due to scattering; standardized to give **1000 W/m²**



Solar Radiation

Air Mass Spectrum AM1.5

- yellow spectrum equals AM0
- **AM1.5 (red curve)** equals the radiation reaching the ground at sea level under an angle of 48.2°

Calculation of AM

$$AM = \frac{1}{\cos(\theta) + 0.50572 \cdot (96.07995 - \theta)^{-1.6364}}$$

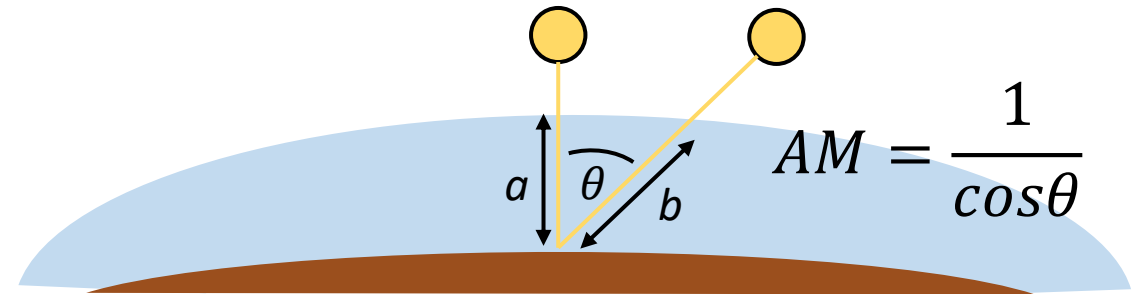
θ : angle of zenith (48.2° for AM 1.5)

$$AM(\theta) = \frac{-R \cdot \cos(\theta) + \sqrt{(R+H)^2 - R^2 \cdot \sin^2(\theta)}}{H}$$

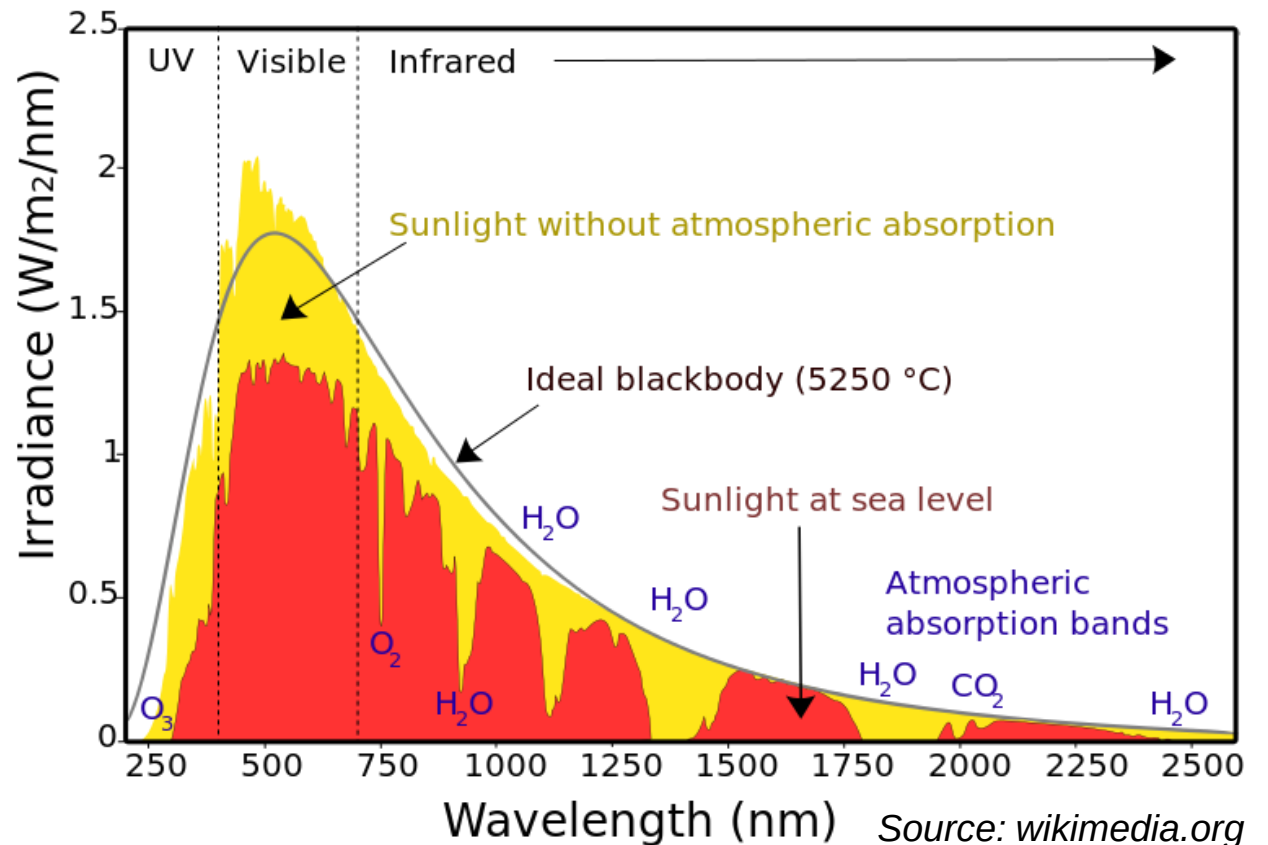
R : radius of earth (6371 km)

H : Scale height (8.5 km)

θ : angle of zenith (48° for AM 1.5)



Spectrum of Solar Radiation (Earth)



Solar Radiation

Solar Intensity Calculation Based on AM

- Intensity of the direct component of the sunlight

$$I_D = I_0 \cdot 0.7^{AM^{0.678}}$$

I_D : intensity on the plane **perpendicular to the radiation** in W/m^2

I_0 is the solar constant or AM0 – the radiation outside earth's atmosphere

- Global intensity I_G includes additional 10 % diffuse radiation:

$$I_G = I_D \cdot 1.1$$

- Taking ground elevations into account:

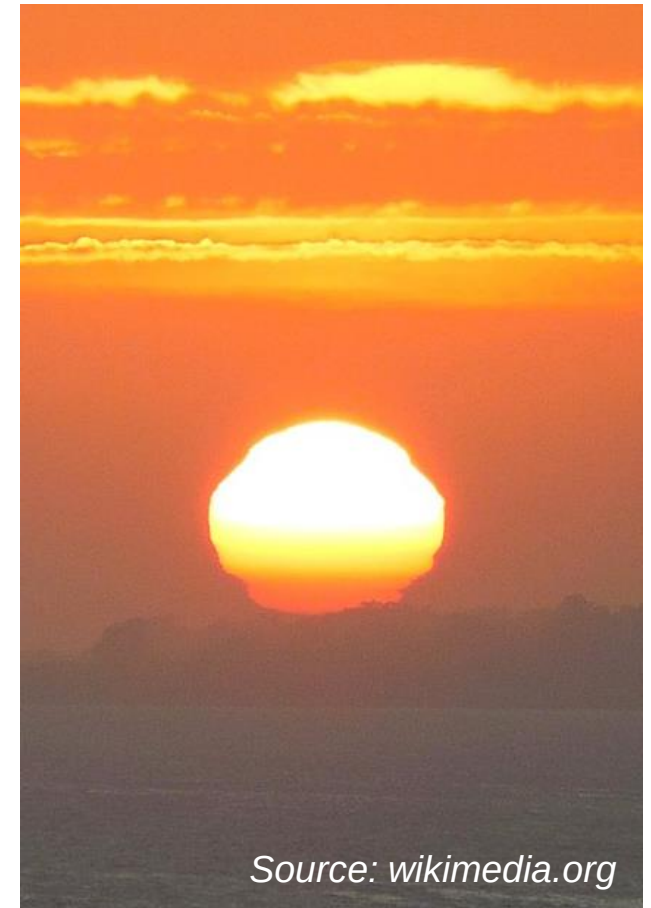
$$I_G = I_0 \cdot 1.1 \cdot \left[(1 - 0.14 \cdot h) \cdot 0.7^{AM^{0.678}} + 0.14 \cdot h \right]$$

h : height above seal level in km (no unit)

- Example: Freiburg, 02.05.2025, 14:00 (48°N 7.85°E, 278 m)

$$\theta = 35.87^\circ \quad AM = 1.23 \Rightarrow I_G = 1017.21 \text{ W/m}^2$$

- Usefull tool: www.pvlighthouse.com.au -> SOLAR SPECTRUM CALCULATOR



Source: [wikimedia.org](https://commons.wikimedia.org/wiki/File:Sunset_in_Freiburg.jpg)

Solar Radiation

Units and Relations

- Photon-Energy: $E_Q = h \cdot f$ and $f = \frac{c}{\lambda}$ (e.g. $2-5 \cdot 10^{-19} J$ for visible light)

- Photon Flux Φ or Solar Radiant flux $\Phi = \frac{\partial E}{\partial t}$ in W

- Spectral irradiance or spectral flux density

$$F(\lambda) = \frac{\Phi}{\Delta\lambda \cdot A} \text{ in } W/(m^2nm) \text{ or } W/m^3$$

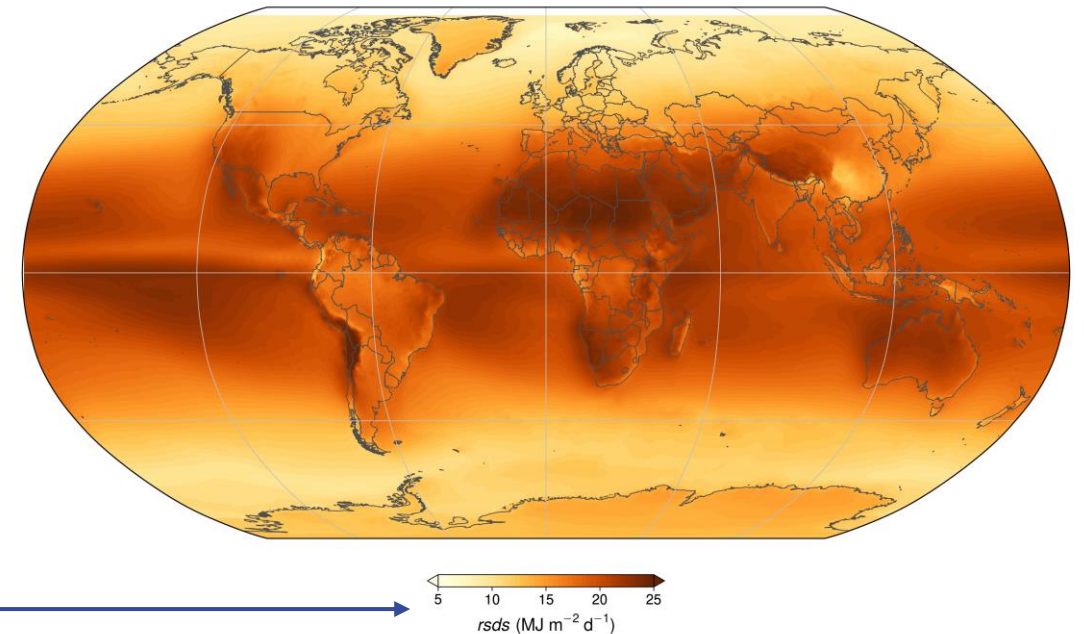
- **Irradiance** or radiant power density

$$H = \int_0^\infty F(\lambda) d\lambda \text{ in } W/m^2 \text{ (flux density)}$$

- Radiance or radiant flux $L_{e,\Omega}$ in $W/(m^2sr)$ (sr: steradian)

- Cumulative (daily) irradiance (e.g. outdoor, European location facing due South at a 45° elevation will receive $1000-5000 \text{ Wh}/m^2/\text{day} = 3.6-18 \text{ MJ}/m^2/\text{d}$)

World map of surface downwelling solar radiation for the period 1981-2010 based on the CHELSA-BIOCLIM+ data set



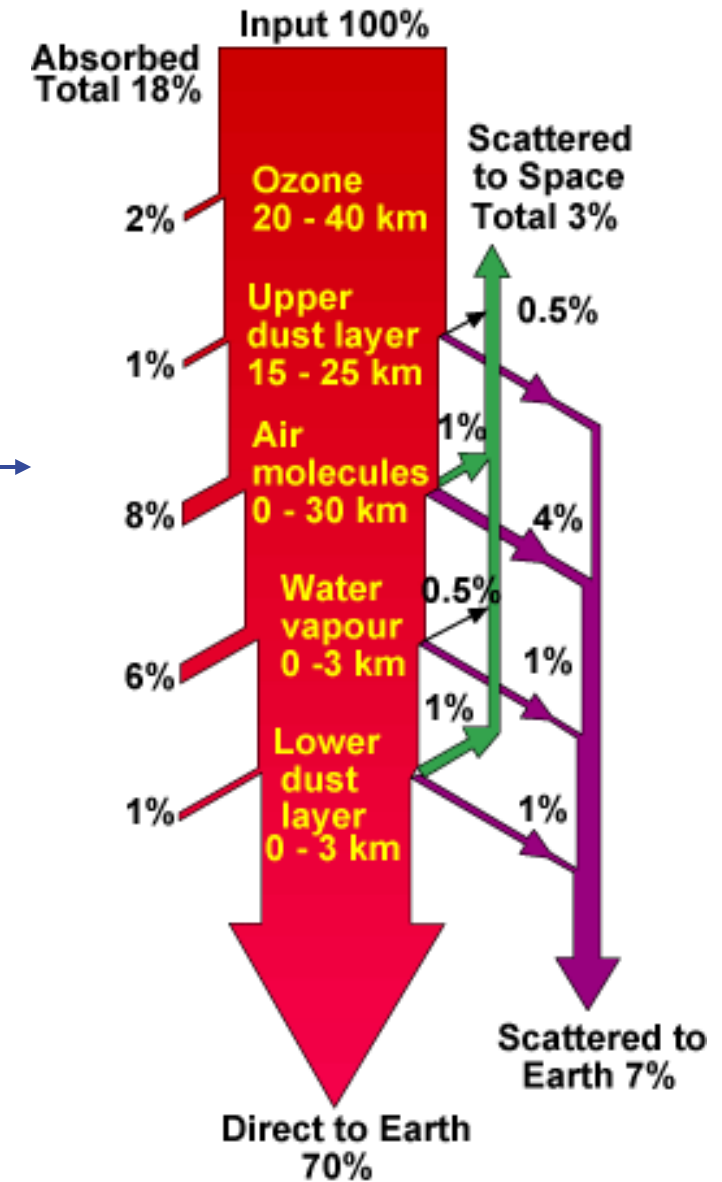
Source: C. Honsberg, S. Bowden, pveducation.org, wikimedia.org and J. F. Randall, PhD Thesis, EPFL, Lausanne, 2003

Solar Radiation

Available Irradiance

Available irradiance on the surface depends on

- Atmospheric absorption and scattering
 - Rayleigh scattering by molecules, molecular absorption spectra
 - Mie scattering by dust particles, aerosols etc. (clouds, air pollution, weather, ...)
- Latitude and location on earth's surface
- Time of the year and time of day



Source: C. Honsberg, S. Bowden, pveducation.org

Solar Radiation

Various Angles

Local Solar Time (LST)

- At midday the sun is highest in sky: can be calculated from the local time LT using correction factors
 - Axis tilt and eccentricity of earth's orbit $EoT = \pm 15 \text{ min}$
 - Longitude-difference to Local Standard Time Meridian (LSTM): usually $\pm 30 \text{ min}$ (but summer time)
Freiburg (7.85°E): $LSTM = 30^\circ\text{E} \Rightarrow -88,6 \text{ min}$
- Sun moves 15° per h: Hour angle $HRA = 15^\circ \cdot (LST - 12)$

Declination Angle δ

- Varies with the season: $\pm 23.45^\circ$ (0° in spring and fall)
- $\delta = 23.45^\circ \cdot \sin \left[\frac{360}{365} \cdot (d - 81) \right]$ with d : days since the start of the year

Source: C. Honsberg, S. Bowden, pveducation.org

Solar Radiation

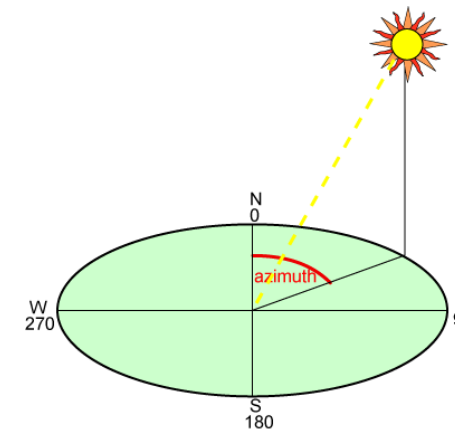
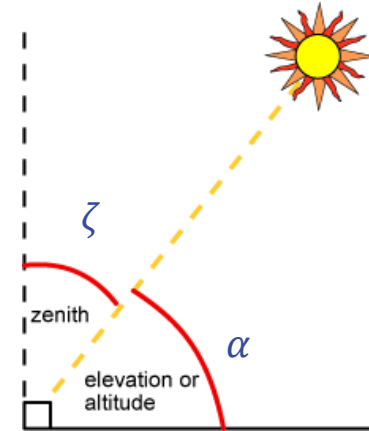
Various Angles

Elevation Angle α

- Angular height of the sun in the sky: usually between 0° (sunrise and sunset) and 90° (directly overhead)
- Elevation angle at noon: $\alpha = 90^\circ + \varphi - \delta$ (if $\alpha > 90^\circ$: $180^\circ - \alpha$) with φ : Latitude of the location (Freiburg: $48^\circ\text{N} \rightarrow +48^\circ$)
- Elevation angle $\alpha = \sin^{-1}[\sin \delta \cdot \sin \varphi + \cos \delta \cdot \cos \varphi \cdot \cos(HRA)]$

Azimuth Angle

- (Compass) direction from which the sun light comes
- $Azimuth^* = \cos^{-1} \left[\frac{\sin \delta \cdot \cos \varphi - \cos \delta \cdot \sin \varphi \cdot \cos(HRA)}{\cos \alpha} \right]$
* for $HRA < 0$; for $HRA > 0$: $360^\circ - Azimuth$



Source: C. Honsberg, S. Bowden, pveducation.org

Solar Radiation

Sun Position and Module Tilt

- Especially important for concentrator modules following the sun's path
- Calculation of incident radiation on tilted surfaces

$$S_h = S_i \cdot \sin \alpha$$

$$S_m = S_i \cdot \sin(\alpha + \beta)$$

with β : module angle

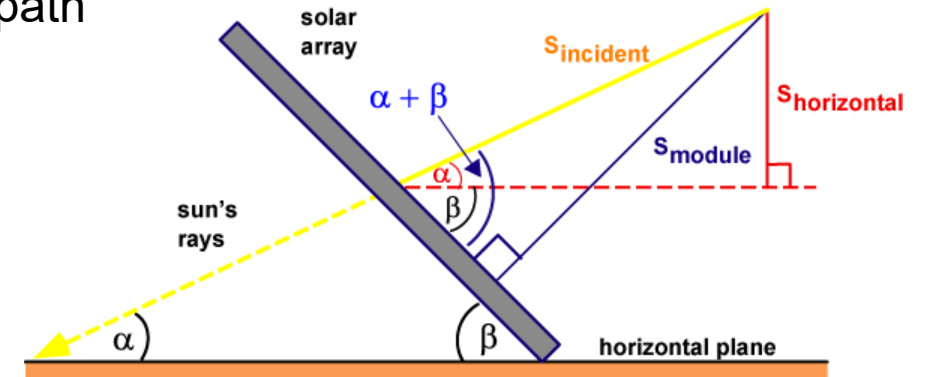
$$\Rightarrow S_m = \frac{S_h \cdot \sin(\alpha + \beta)}{\sin \alpha}$$

Solar radiation on...

S_m : module surface

S_h : a horizontal surface

S_i : direct



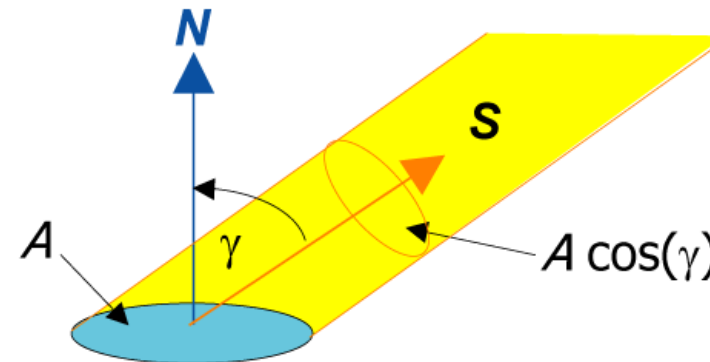
module oriented towards equator

- Arbitrary Module Orientation:

$$S_m = S_i \cdot [\cos \alpha \sin \beta \cos(\psi - \Theta) + \sin \alpha \cos \beta]$$

with ψ and Θ : Azimuth angles of module and sun

- Using Vectors: dot product between incident light and module normal $S_m = S_i \cdot \cos \gamma = S_i \cdot \mathbf{S} \cdot \mathbf{N}$

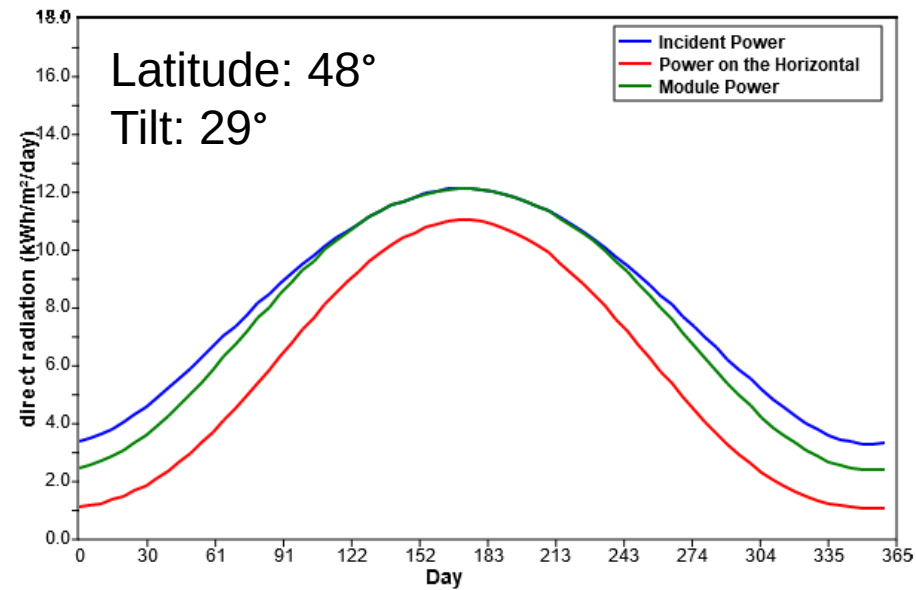
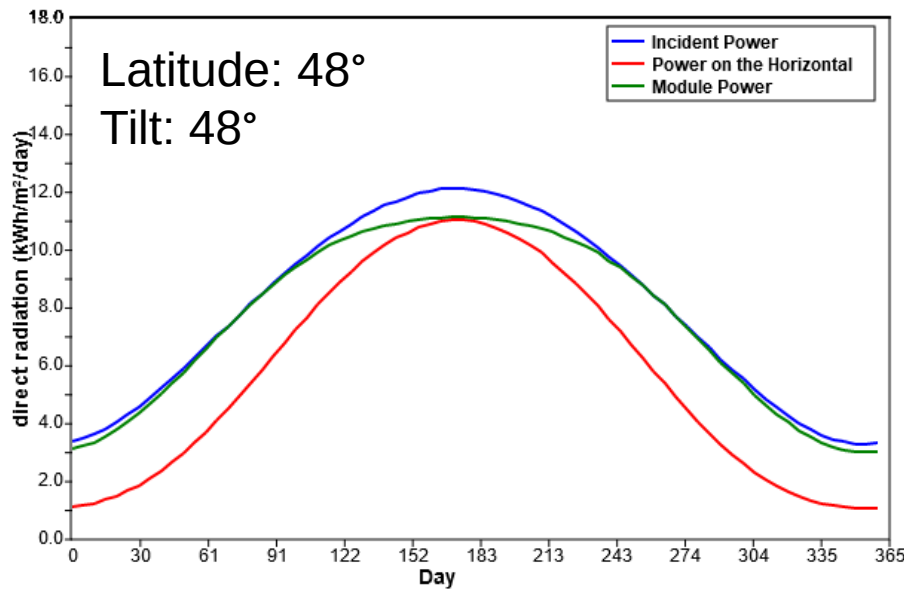


Source: C. Honsberg, S. Bowden, pveducation.org

Solar Radiation

Optimal Module Tilt

- Usually determined by the latitude for fixed position modules
- For seasonal applications a different tilt angle might be beneficial
- Solar insolation for different module tilts:

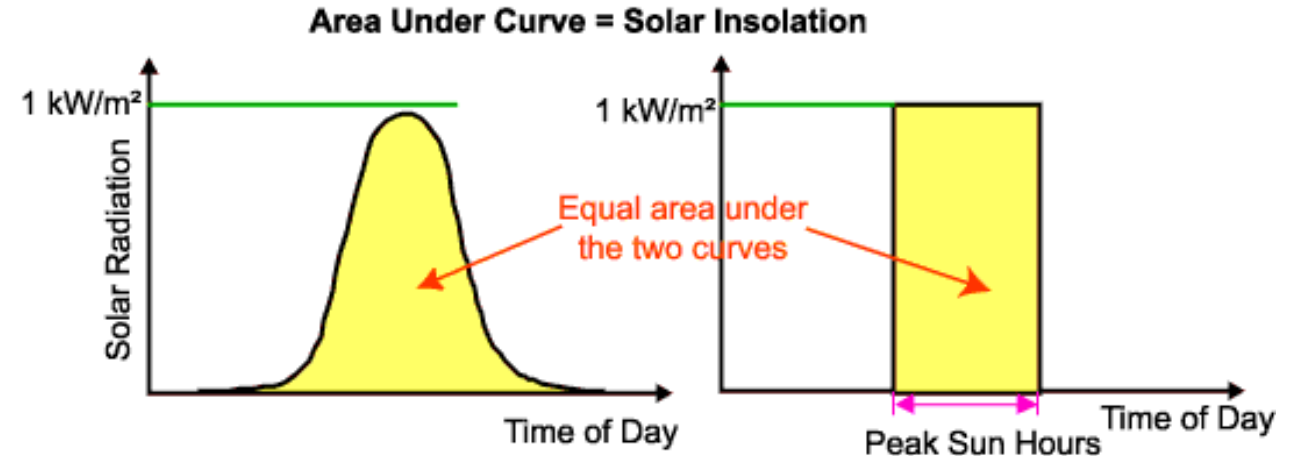


Source: C. Honsberg, S. Bowden, pveducation.org

Solar Radiation

Solar Irradiance and Solar Insolation

- **Solar insolation** is the total amount of solar energy received during a specified time period (integral of the solar irradiance) in $\text{kWh/m}^2/\text{day}$ ($\rightarrow \text{W/m}^2$)



- **Solar irradiance is measured, solar insolation is used for PV-system design**
- Radiation measurement
 - Pyranometer: global irradiance (on a horizontal surface)
 - Pyrhelimeter: measurement of direct radiation only (sun-following)
 - Sunshine recorders: measures hours of sunshine above a certain threshold (200 mW/cm^2)



Source: C. Honsberg, S. Bowden, pveducation.org, wikimedia.org

Solar Radiation

Solar Insolation Calculation using TMY Data

- Data available in TMY (typical meteorological year) datasets for a given location
- Tables contain hourly **Global Irradiance** (I_G) on a horizontal surface as well as **Tracking Beam Irradiation** (I_D) data
- Diffuse Irradiation I_{Diff} can be calculated roughly by:
- Total Insolation (G) received by any fixed module in arbitrary orientation is composed of the direct part (B) as well as the diffuse irradiation (D):

$$I_{Diff} = I_G - I_D \cdot \sin \alpha$$

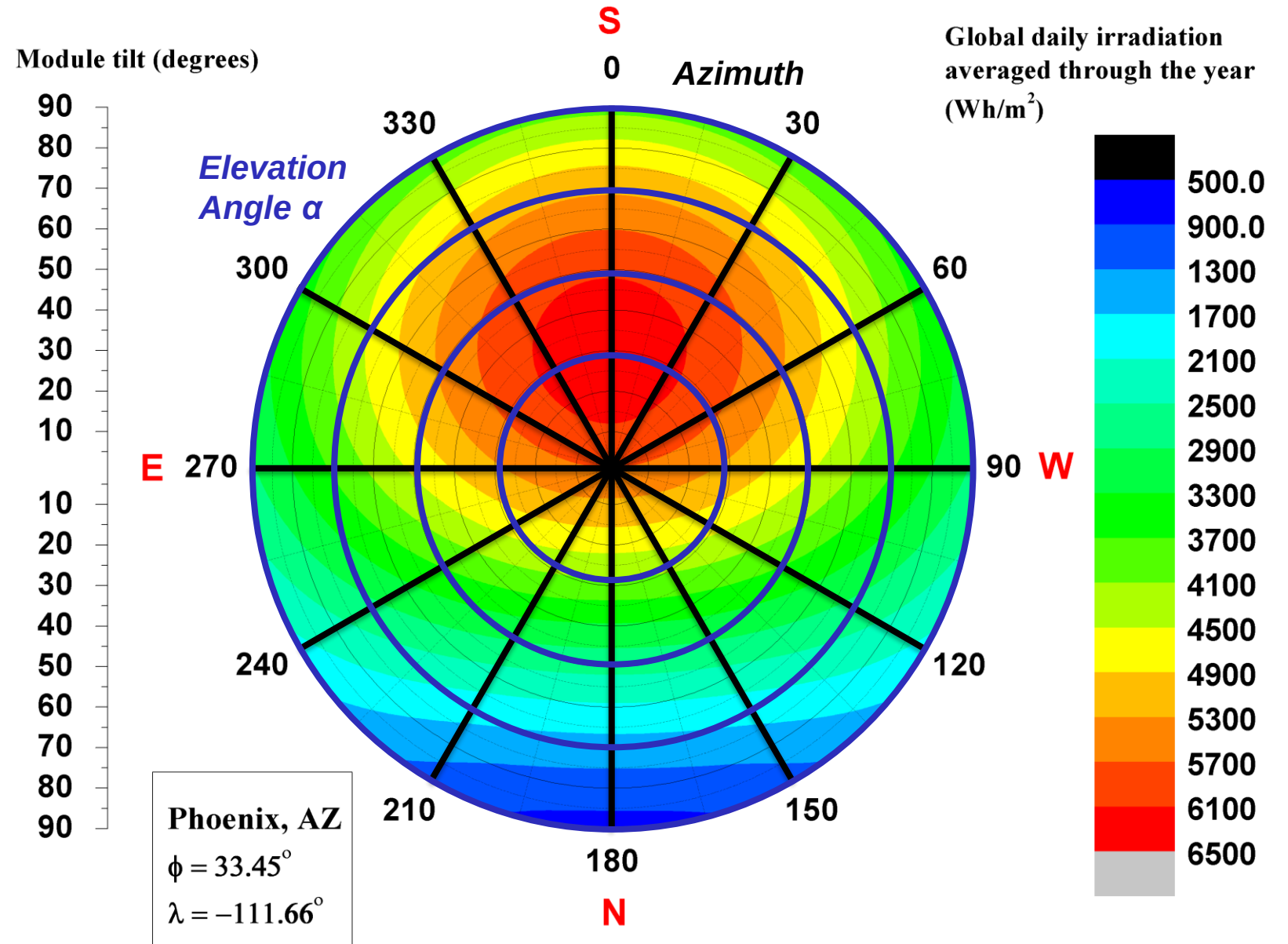
$$G = B + D$$

$$B = I_D \cdot [\sin \delta \sin \varphi \cos \beta - \\ - \sin \delta \cos \varphi \sin \beta \cos \psi + \cos \delta \cos \varphi \cos \beta \cos(HRA) + \\ + \cos \delta \sin \varphi \sin \beta \cos \psi \cos(HRA) + \cos \delta \sin \psi \sin(HRA) \sin \beta]$$

$$D = I_{Diff} \left(\frac{180 - \beta}{180} \right)$$

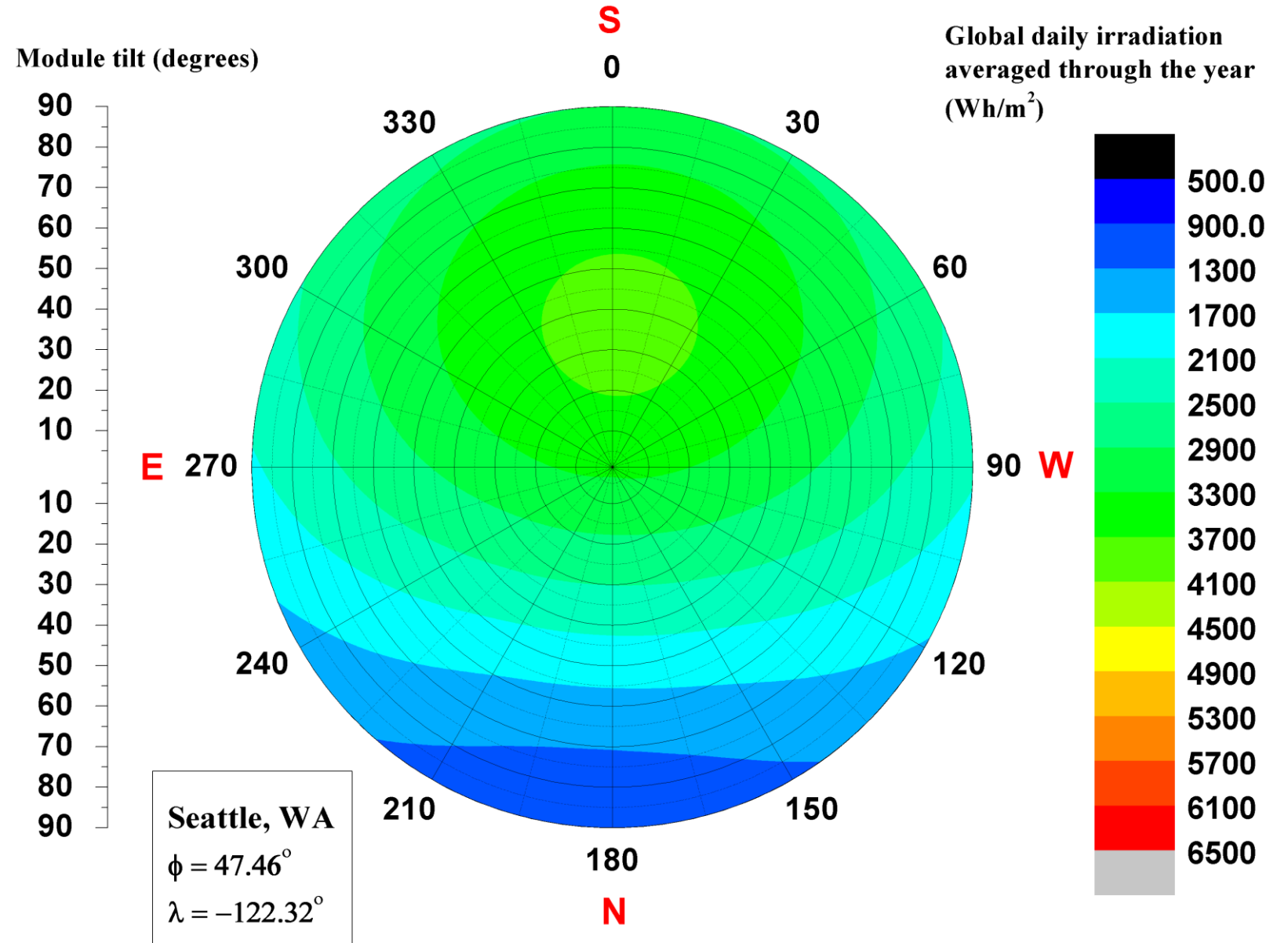
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Solar Radiation Examples



Source: C. Honsberg, S. Bowden, pveducation.org, wikimedia.org

Solar Radiation Examples

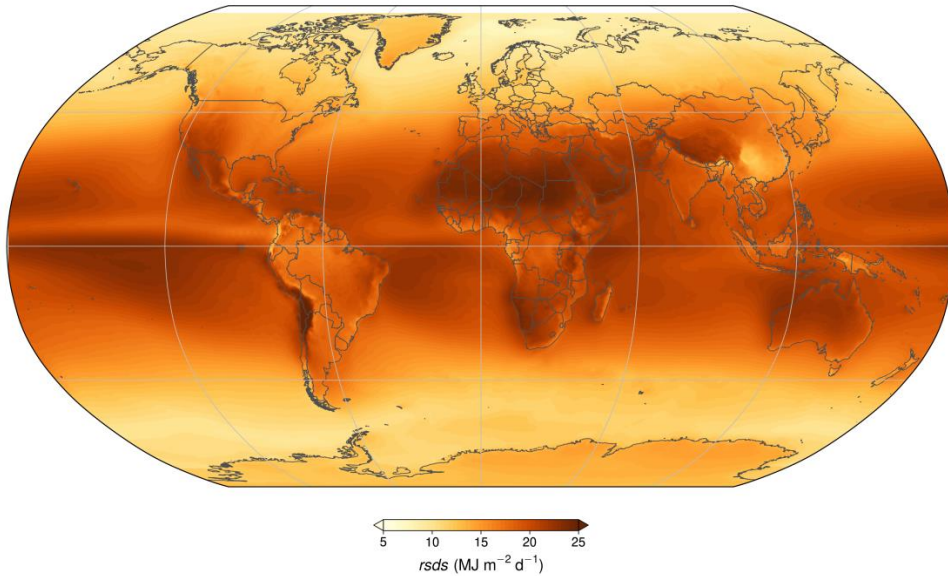


Solar Radiation

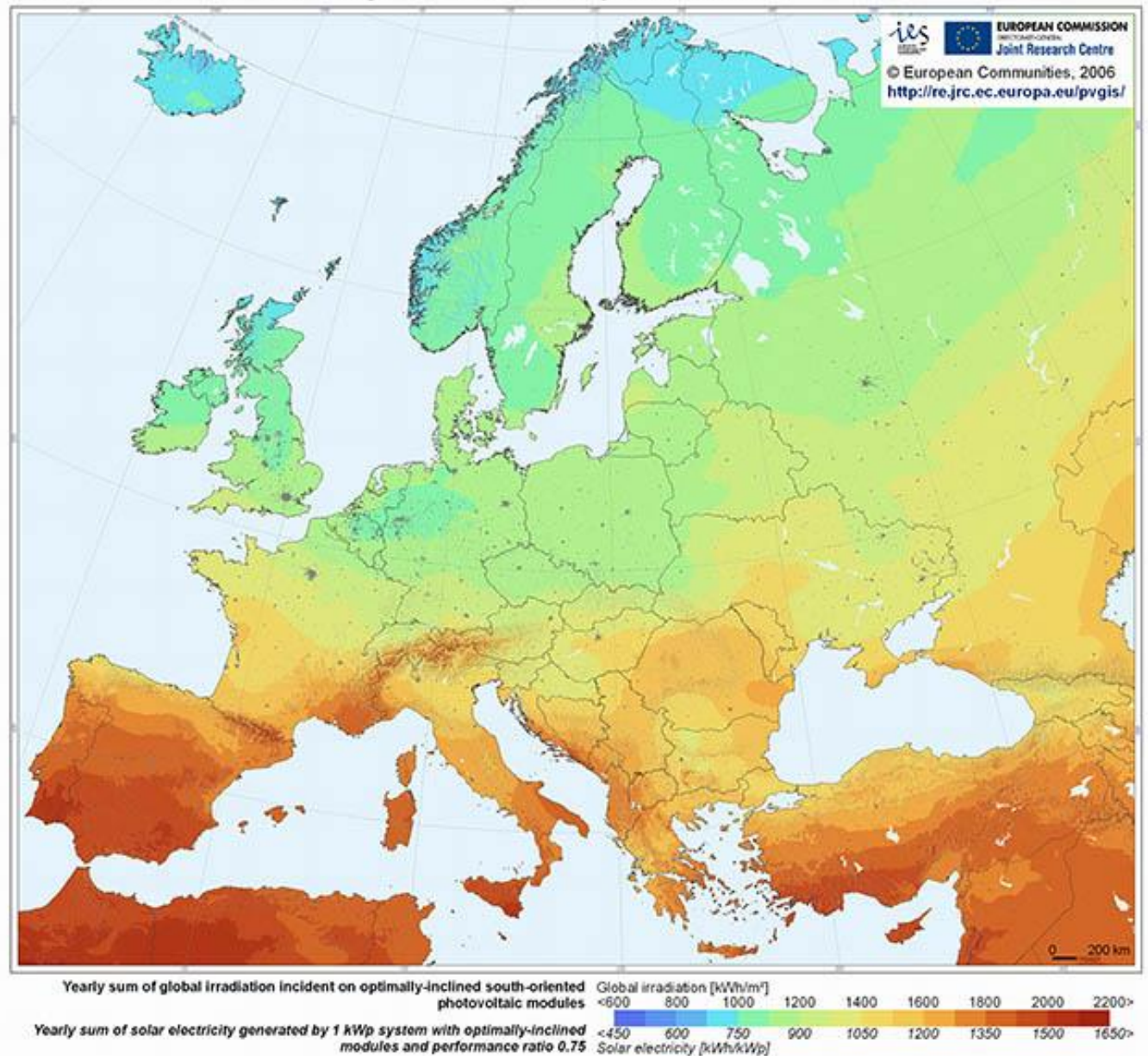
Global Solar Irradiance Maps

- Useful to determine average effects of e.g. cloud cover

World map of surface downwelling solar radiation for the period 1981-2010 based on the CHELSA-BIOCLIM+ data set



Photovoltaic Solar Electricity Potential in European Countries



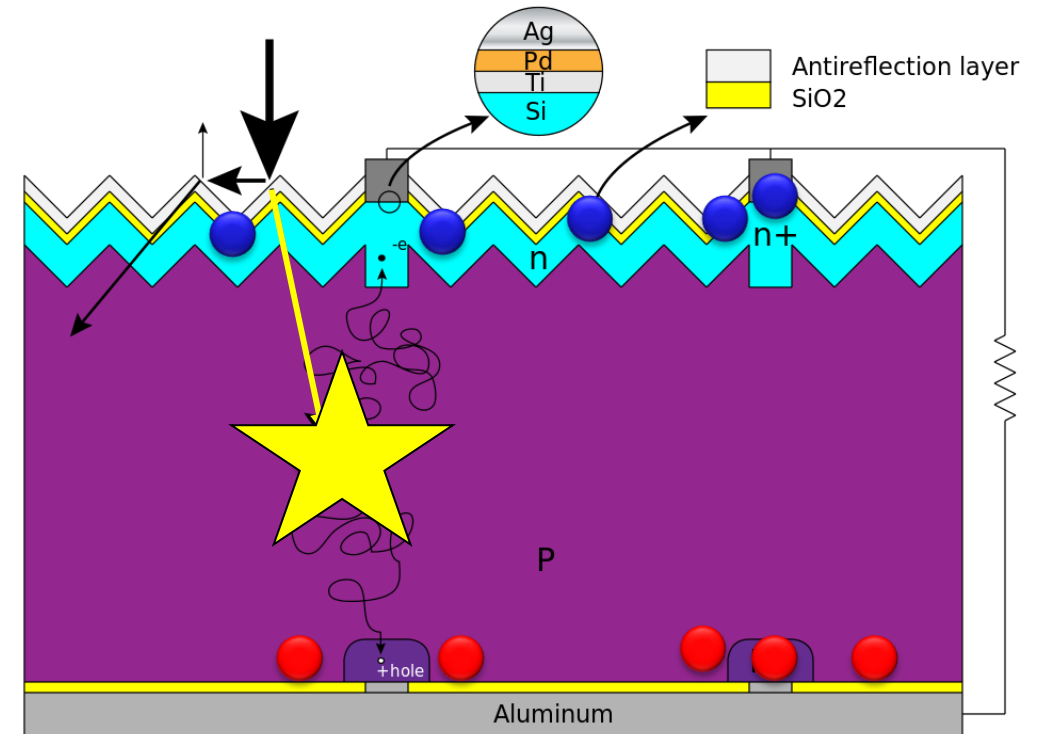
Photovoltaic Cells



Photovoltaic Cells

Working Principle

- Light induced e^- - h^+ -pair generation
→ Recombination or ↓
- Carrier collection at the electrodes to generate a current
- Generation of a voltage across the solar cell (PV-Effect)
- Dissipation of power in the load and parasitic resistances

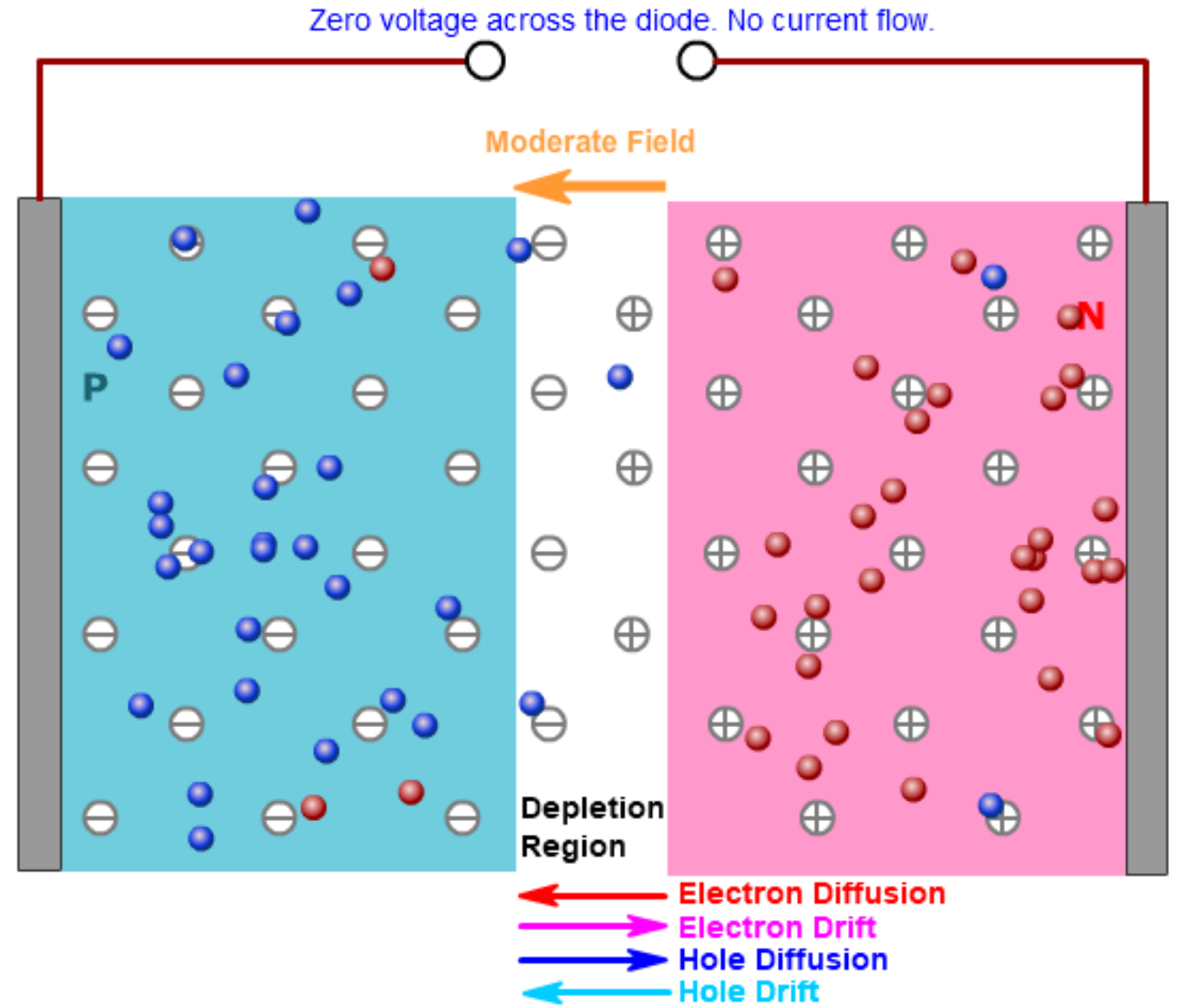


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Modes of Operation

- Equilibrium conditions:
no light, no voltage, no current

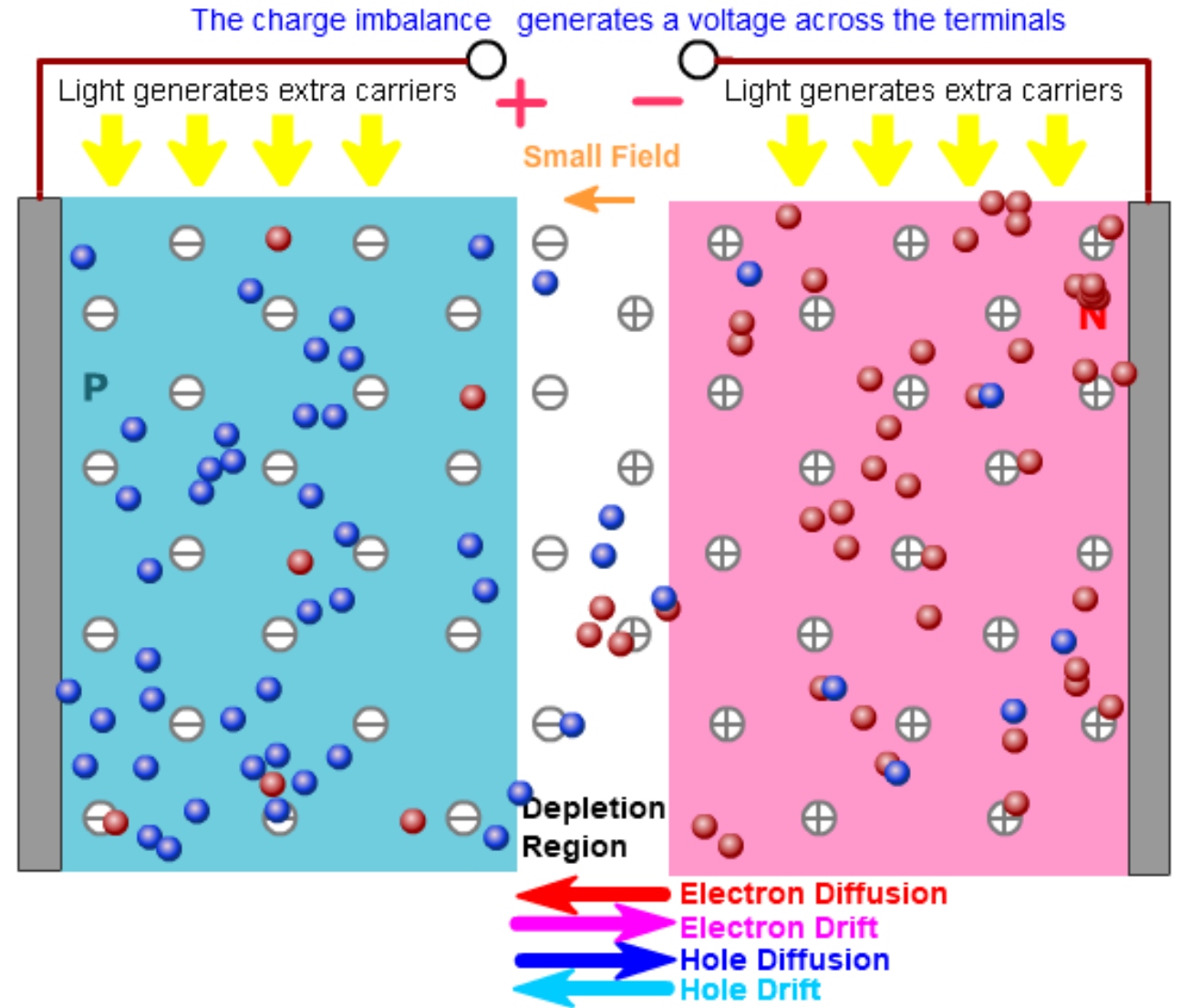


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Modes of Operation

- Equilibrium conditions:
no light, no voltage, no current
- Open circuit conditions:
 - additional CC are generated
 - V_{OC} across the terminals
 - E field opposes the internal field
 - smaller effective field in the DR

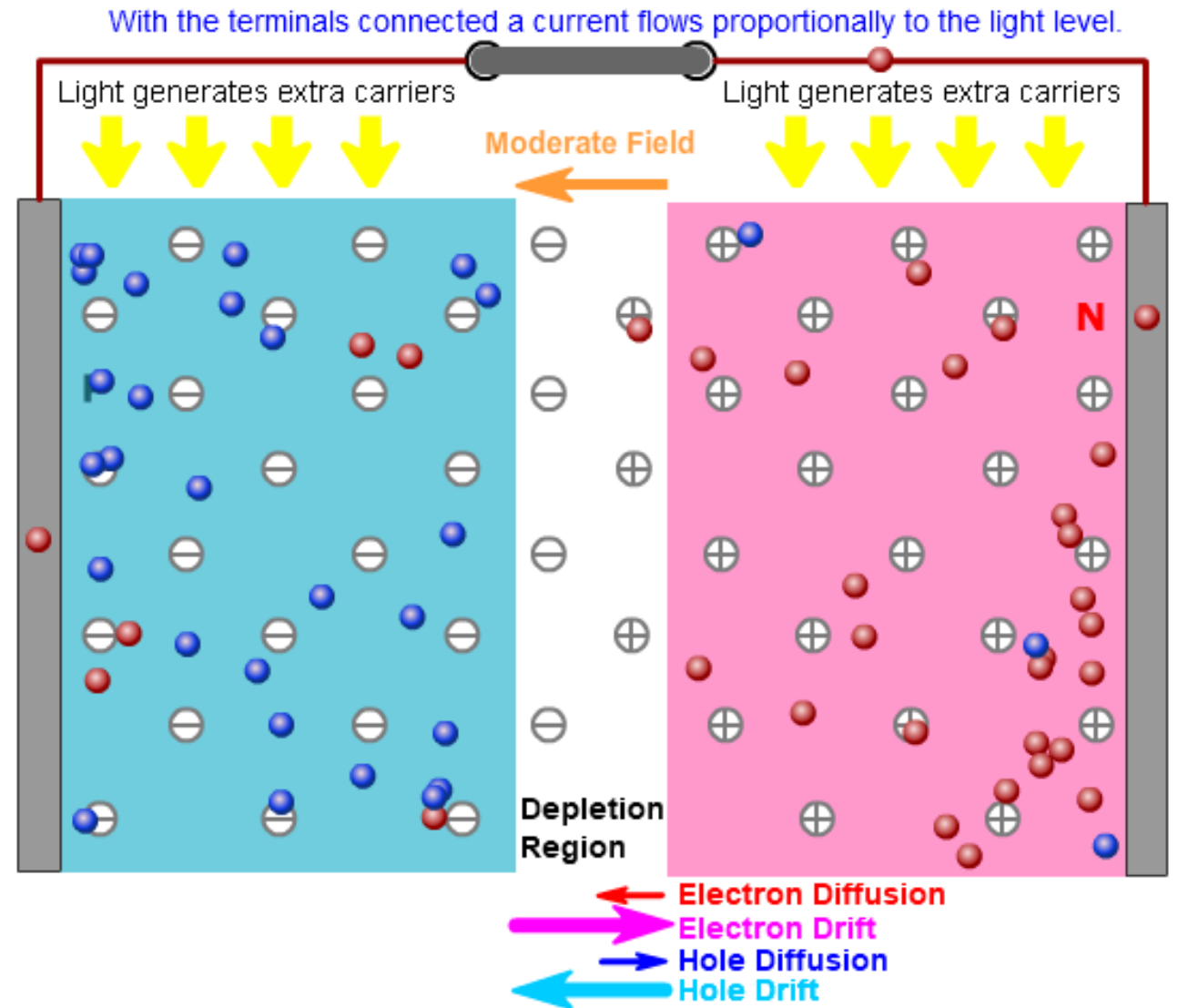


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Modes of Operation

- Equilibrium conditions:
no light, no voltage, no current
- Open circuit conditions:
 - additional CC are generated
 - V_{OC} across the terminals
→ E field opposes the internal field
→ smaller effective field in the DR
- Short circuit conditions:
 - I_{SC} is allowed to flow through the external connections
 - I_{SC} is proportional to the irradiance

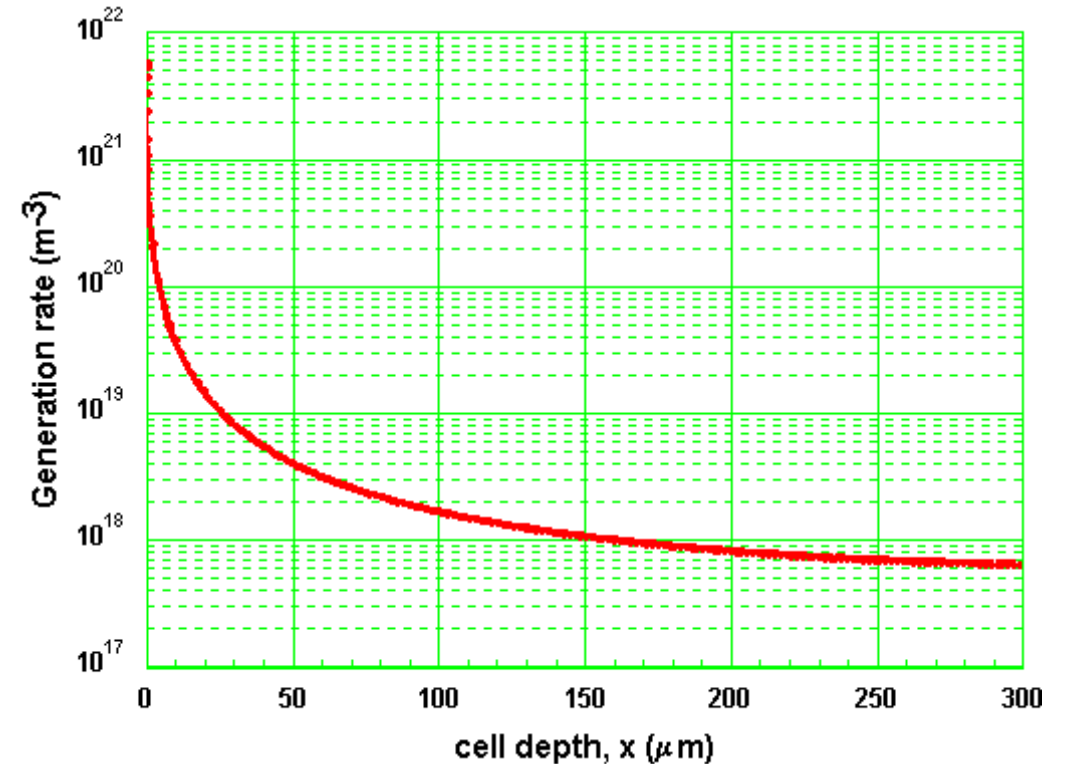
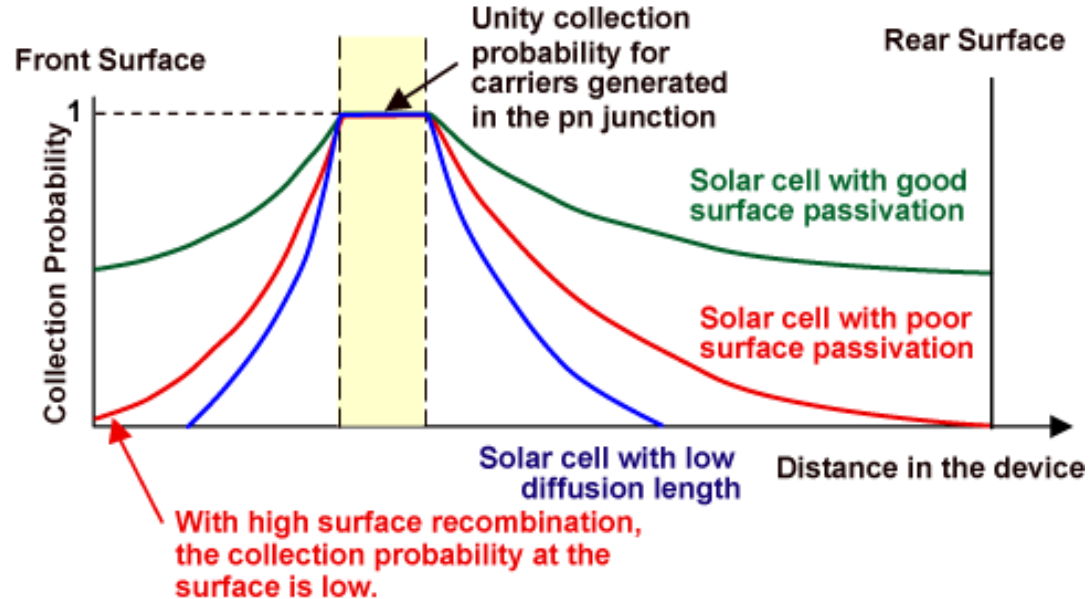


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Collection Probability

- Main influences
 - Charge carrier travel distance compared to
 - Charge carrier diffusion length
 - Surface properties of the solar cell



- General trend of the collection probability and generation rate in silicon

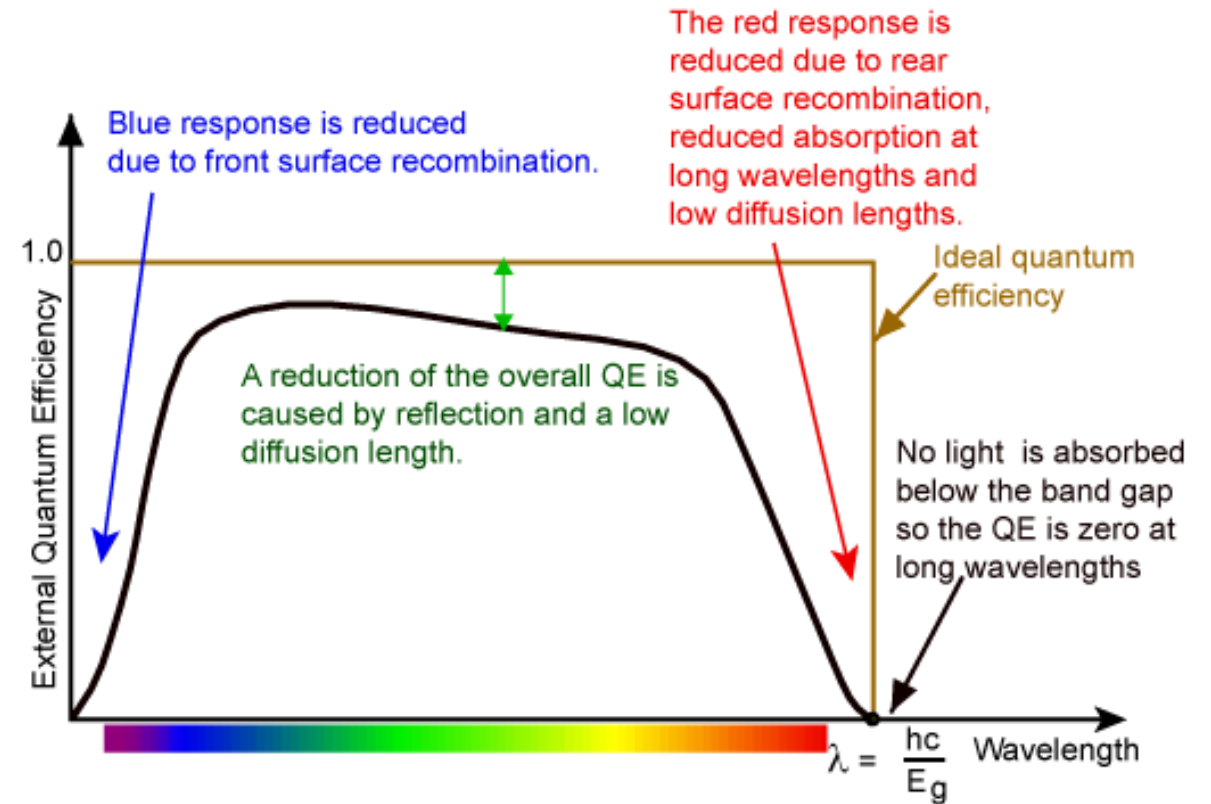
Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Quantum Efficiency QE

$$QE = \frac{\# \text{ electrons collected}}{\# \text{ incident photons}}$$

- QE is the collection probability due the generation profile of a single wavelength (seen earlier), integrated over the device thickness and normalized to the number of incident photons
- External QE – including reflection and transmission
- Internal QE – reflection and transmission are omitted



Source: C. Honsberg, S. Bowden, pveducation.org

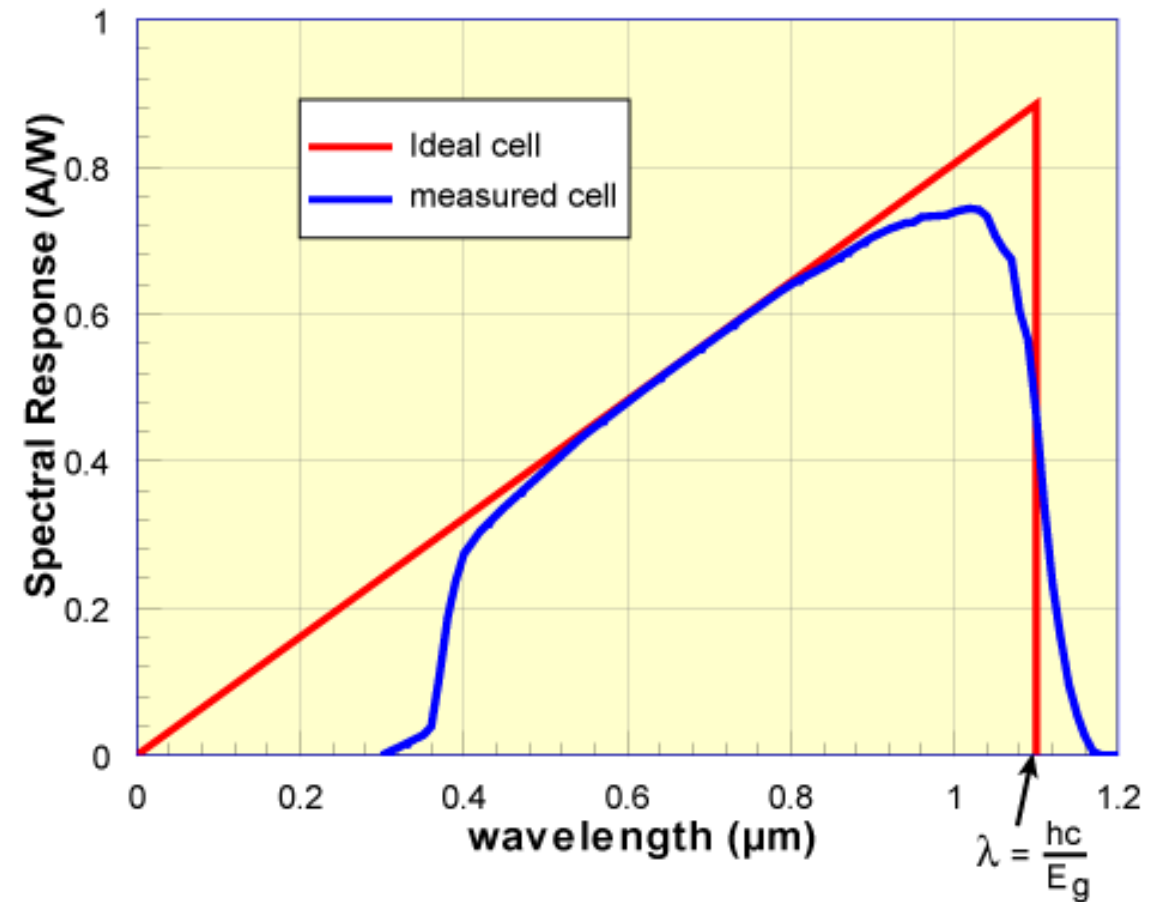
Photovoltaic Cells

Spectral Response SR

- SR is the ratio of the current generated to the power incident on the solar cell

$$SR(\lambda) = \frac{I_{sc}}{P_{in}} = \frac{q\lambda}{hc} QE \text{ in A/W}$$

- SR can be measured and from it, the QE can be calculated

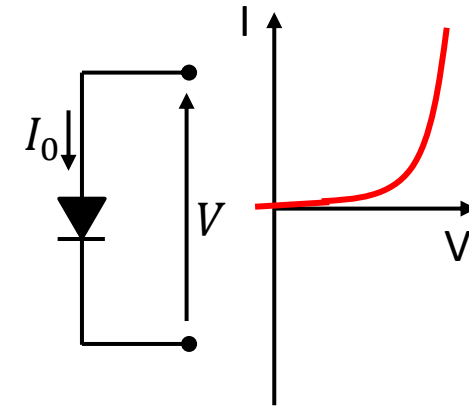


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

IV Curve

- Diode equation without illumination
 $I = I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right)$ with n: ideality factor
→ dark current

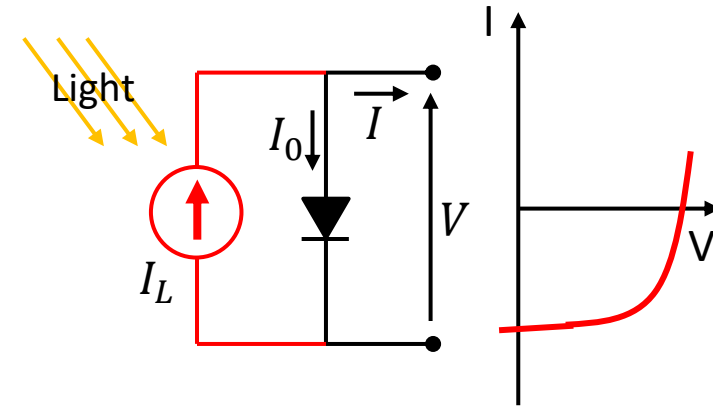


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

IV Curve

- Diode equation without illumination
 $I = I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right)$ with n: ideality factor
→ dark current
- Radiation results in an additional light generated current I_L
 $I = I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right) - I_L$

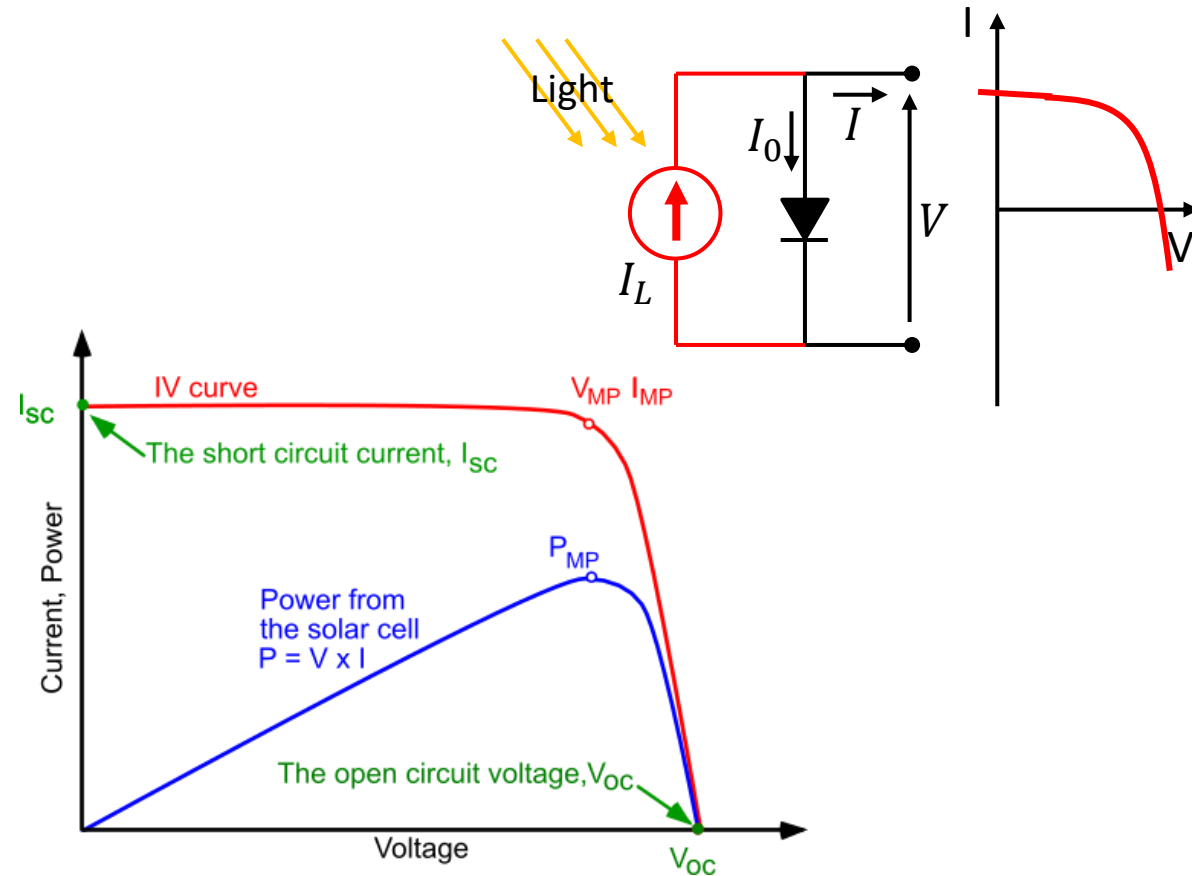


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Photovoltaic Cells

IV Curve

- Diode equation without illumination
 $I = I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right)$ with n: ideality factor
→ dark current
- Radiation results in an additional light generated current I_L
 $I = I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right) - I_L$
- Inverting the current axis as the cell generates power → IV Curve
 $I = I_L - I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right)$

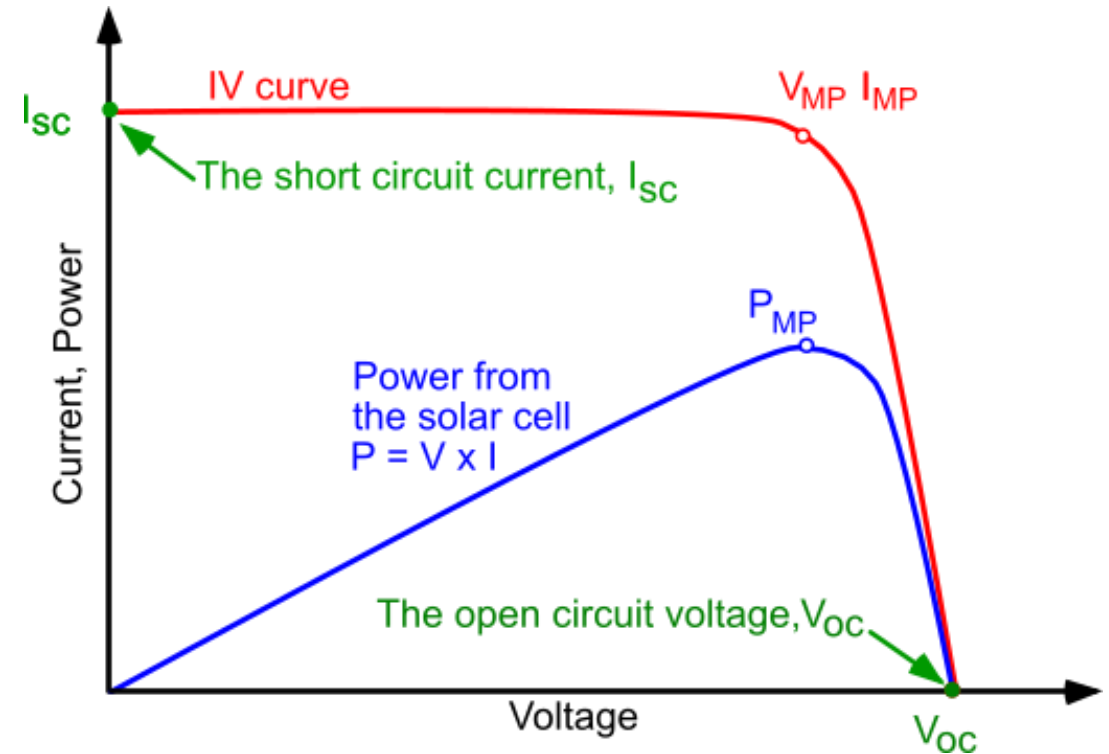


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

IV Curve

- Short Circuit Current I_{SC}
 - I_{SC} is measured and can be used as $I_{SC} = I_L$ except for very high series resistance in the cell
 - Decreases with increasing bandgap
- Open Circuit Voltage V_{OC}
 - $V_{OC} = \frac{nkT}{q} \ln \left(\frac{I_L}{I_0} + 1 \right)$ or $V_{OC} = \frac{kT}{q} \ln \left[\frac{(N_A + \Delta n) \cdot \Delta n}{n_i^2} \right]$
with: N_A : doping concentration; Δn excess and n_i intrinsic CC
 - $V_{OC} \propto$ amount of recombination in the device
 - Increases with increasing bandgap



Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

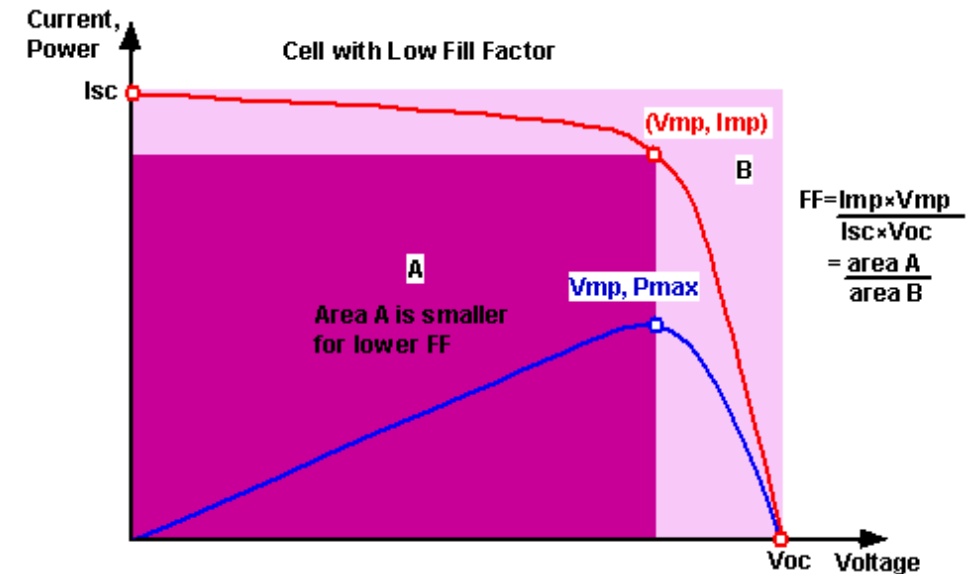
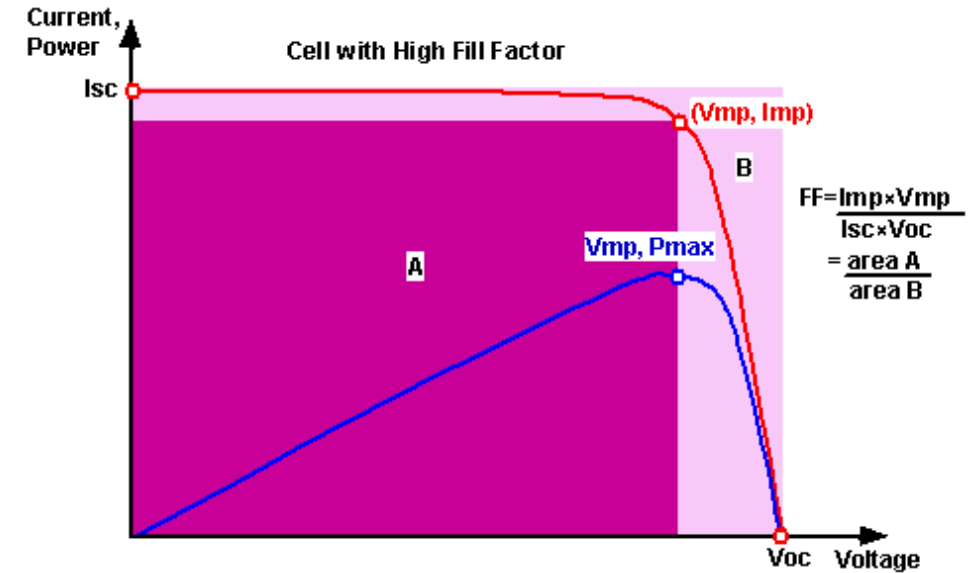
Fill Factor FF

$$FF = \frac{P_{MP}}{V_{OC} \cdot I_{SC}} = \frac{V_{MP} \cdot I_{MP}}{V_{OC} \cdot I_{SC}}$$

- Maximum FF for $\frac{d(IV)}{dV} = 0$

$$V_{MP} = V_{OC} - \frac{nkT}{q} \ln\left(\frac{qV_{MP}}{nkT} + 1\right)$$
 (solvable via Lambert W funktion)
- Empirical expression for FF

$$FF = \frac{v_{OC} - \ln(v_{OC} + 0.72)}{v_{OC} + 1}$$
 with $v_{OC} = \frac{q}{nkT} V_{OC}$
- Highest FF values in the range of 0.85 with some cells approaching 0.90



Source: C. Honsberg, S. Bowden, pveducation.org

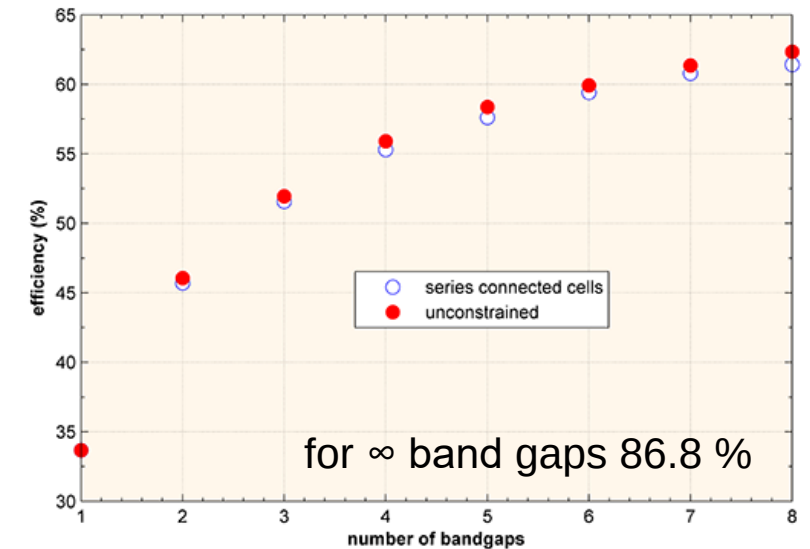
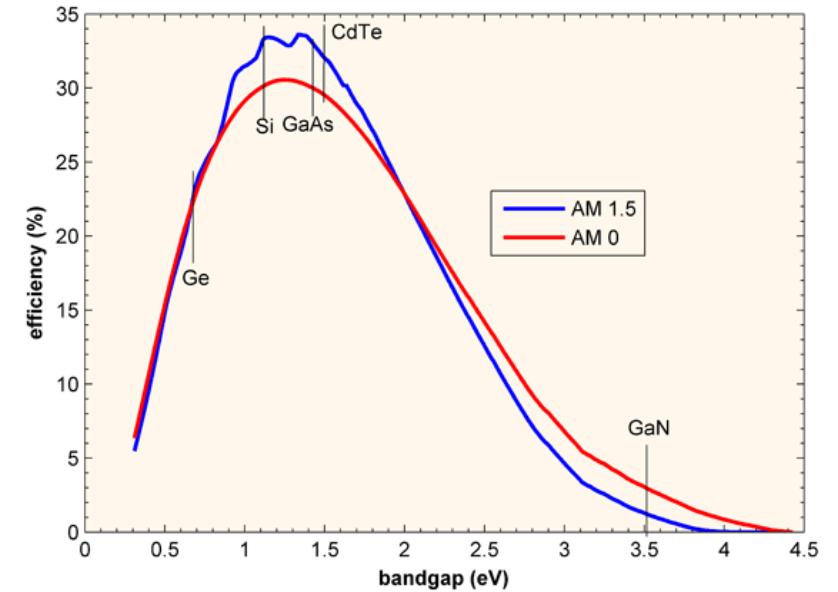
Photovoltaic Cells

Solar Cell Efficiency

- Related to FF, V_{OC}, I_{SC} :

$$\eta = \frac{P_{max}}{P_{in}} = \frac{V_{OC} \cdot I_{SC} \cdot FF}{P_{in}}$$

- P_{in} is usually calculated based on AM1.5 irradiation of 1000 W/m²
→ 10 W for a 10x10 cm² cell
- Maximum theoretical efficiency?
 - Shockley–Queisser limit for single p-n-junction solar cells:
33.7 % @ 1.34 eV
 - SQ limit for Si (1.1 eV) – 32 %
 - Tandem and Multijunction cells as well as concentrator cells outperform this limit

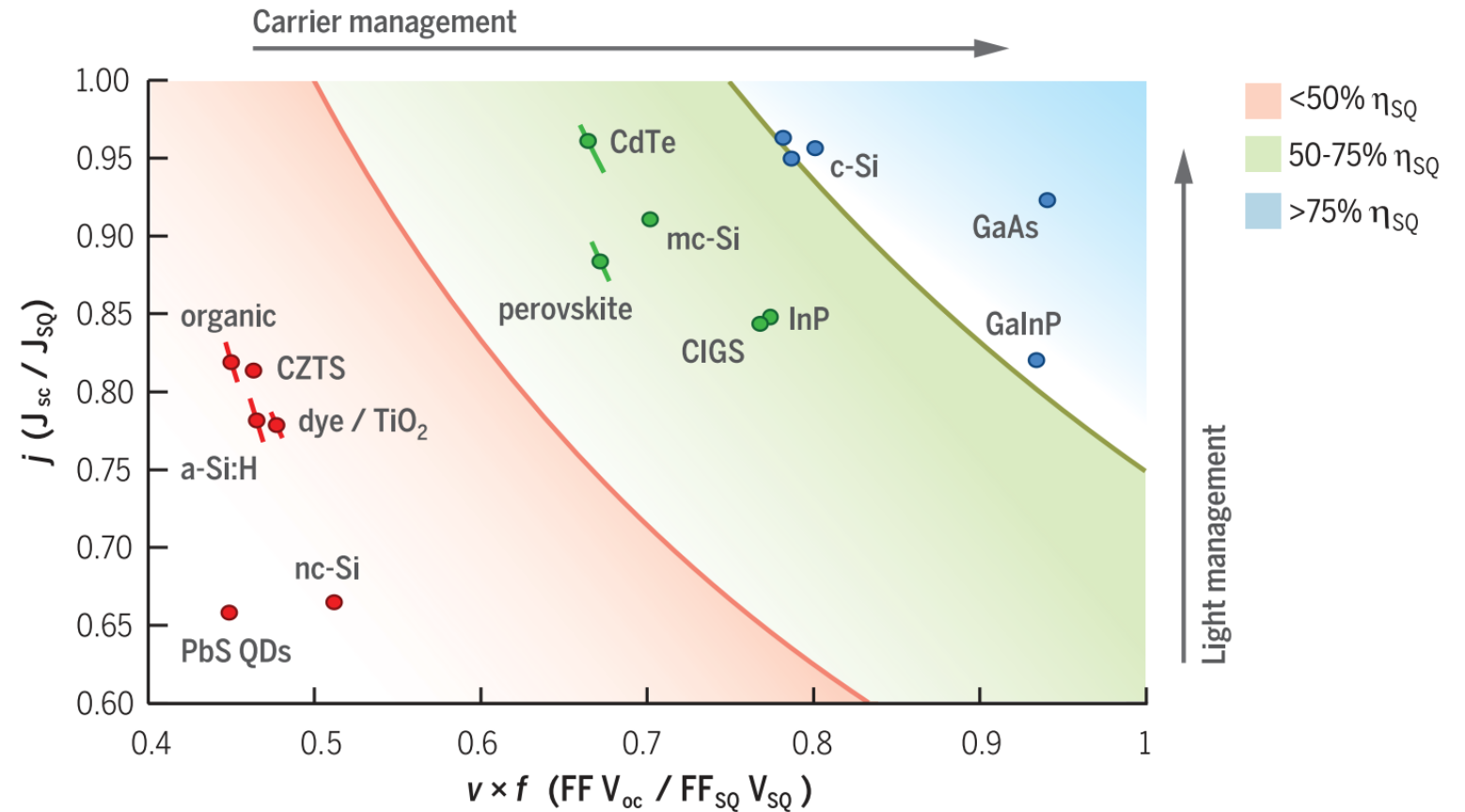


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Photovoltaic Cells

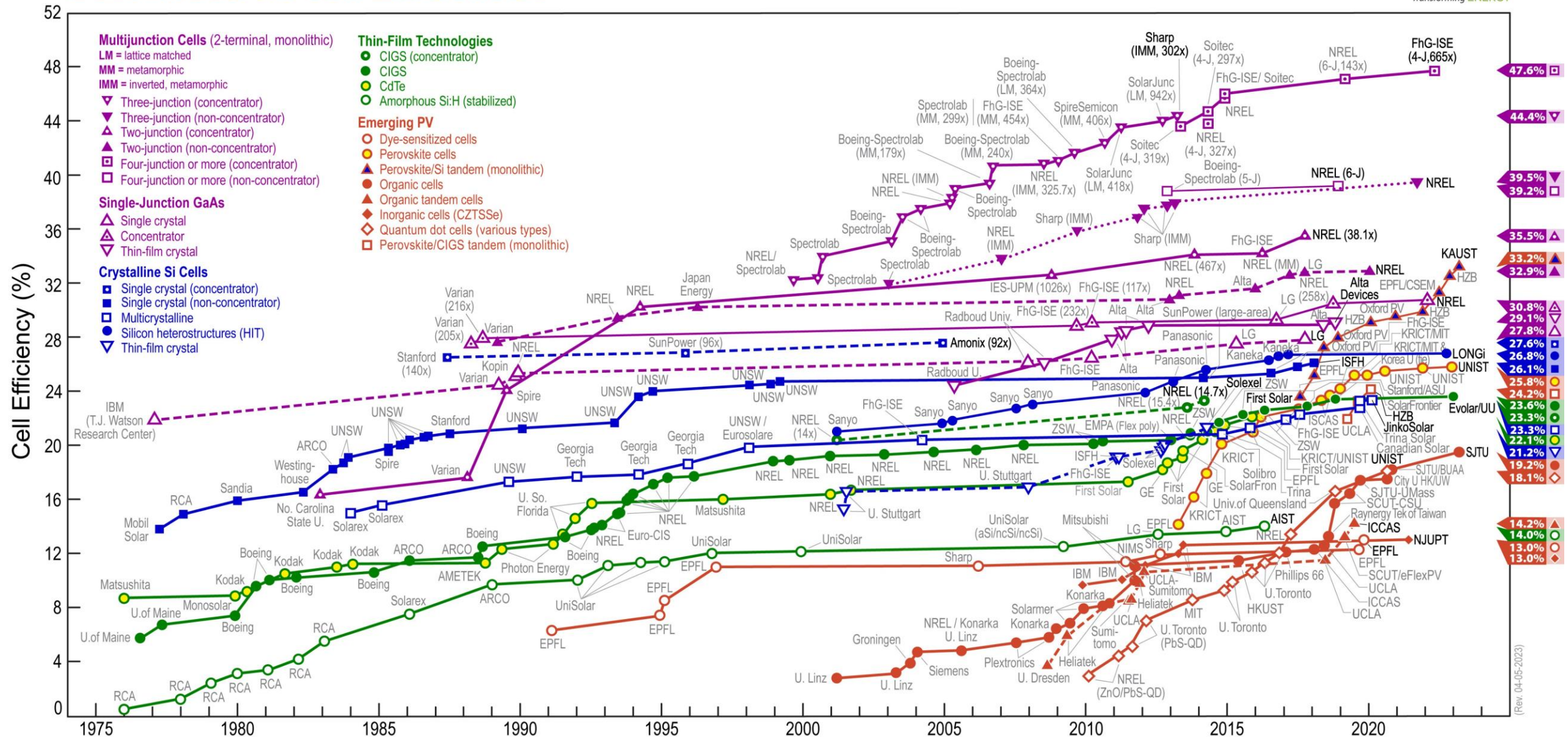
Solar Cell Efficiency

Maximum conversion efficiency
for a single junction cell:
33.7 % @ 1.34 eV



Source: A. Polman et.al. *Science* (New York, N.Y.) **2016**, 352, aad4424.

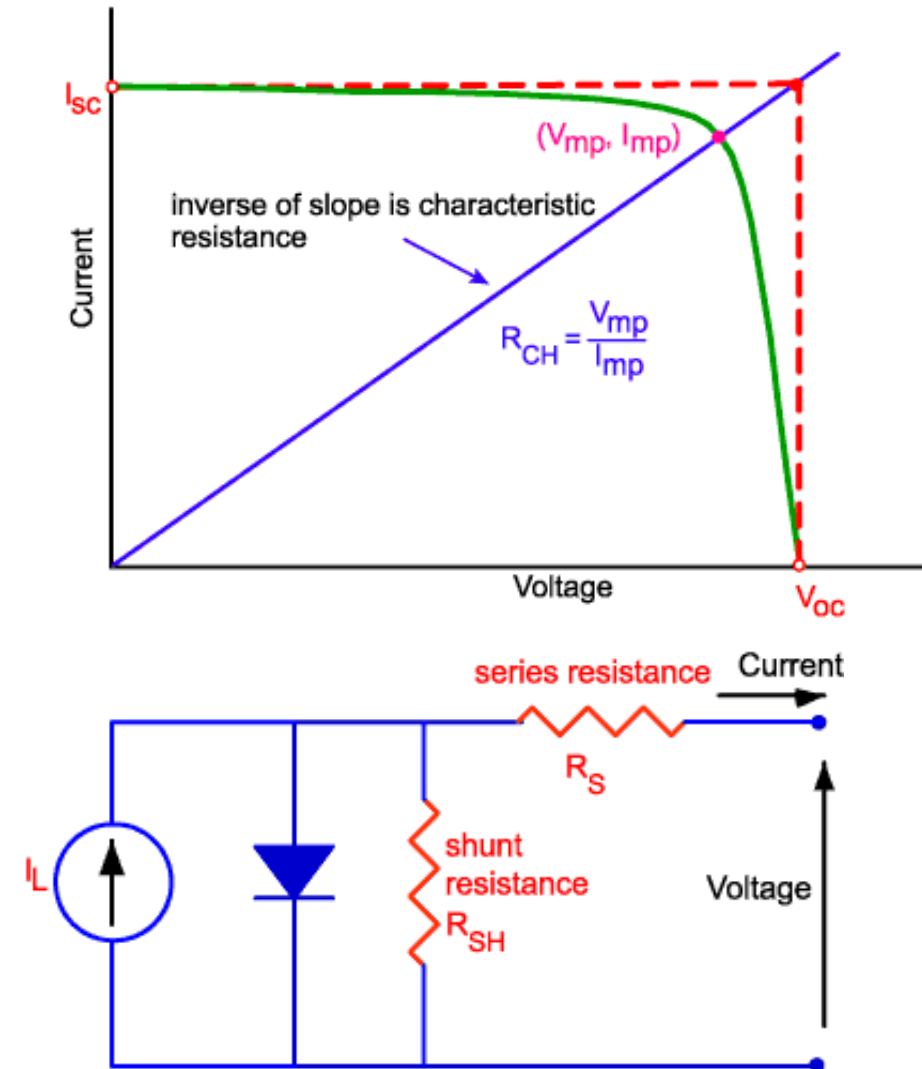
Best Research-Cell Efficiencies



Photovoltaic Cells

Resistances in Solar Cells

- Characteristic resistance R_{CH} at the maximum power point ($R_{Load} = R_{CH}$)
$$R_{CH} = \frac{V_{MP}}{I_{MP}} \approx \frac{V_{OC}}{I_{SC}} \text{ (typ. in the m}\Omega \text{ range)}$$
- Series Resistance R_S (typ. 0.5 – 1,3 Ω/cm^2)
 - Bulk resistance of emitter and base
 - Contact resistance between metal and Si
 - Resistance of the metal electrodes
- Shunt Resistance R_{SH} (typ. > 1000 Ω/cm^2)
 - Leakage currents due to (manufacturing) defects
- Parasitic resistances **reduce FF directly**, but have little effect on V_{OC} and I_{SC}

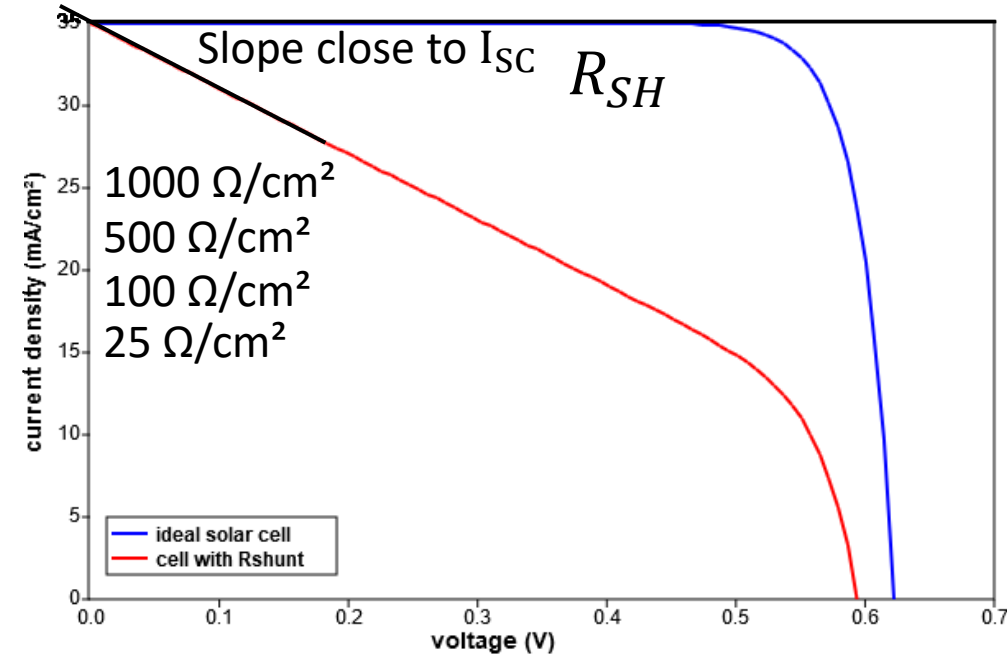
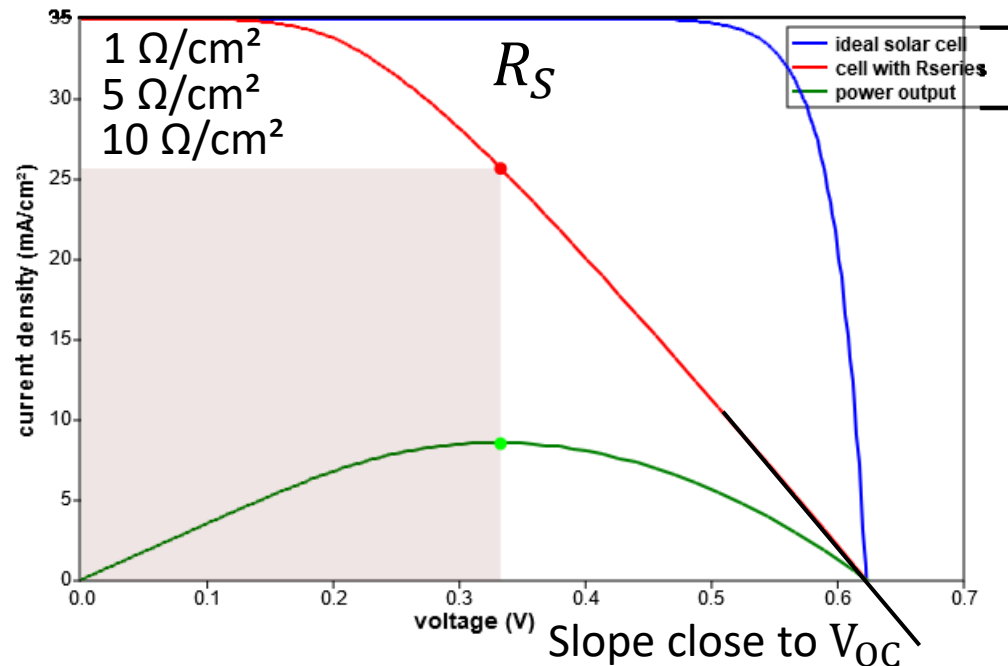


Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Resistances in Solar Cells

	Influence of R_S	Influence of R_{SH}
Solar cell current	$I = I_L - I_0 \cdot \left(e^{\frac{q(V+IR_S)}{nkT}} - 1 \right)$	$I = I_L - I_0 \cdot \left(e^{\frac{qV}{nkT}} - 1 \right) - \frac{V}{R_{SH}}$
Combined	$I = I_L - I_0 \cdot \left(e^{\frac{q(V+IR_S)}{nkT}} - 1 \right) - \frac{V + IR_S}{R_{SH}}$	



Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Parasitic Resistance Impact on FF

- For small R_S and large R_{SH} : $P'_{MP} \approx P_{MP} - P_{Lost}$
- Reduced resistances $r_S = \frac{R_S}{R_{CH}}$ and $r_{SH} = \frac{R_{SH}}{R_{CH}}$
- Empirical equations for the influenced FF_S/FF_{SH} vs. FF_0 (no add. resistance):

$$FF_S = FF_0 \cdot (1 - 1.1r_S) + \frac{r_S^2}{5.4} \text{ and } FF_{SH} = FF_0 \cdot \left(1 - \frac{V_{OC}+0.7}{V_{OC}} \cdot \frac{FF_0}{r_{SH}}\right)$$

- The combination results an empirical equation for FF

$$FF = FF_0 \cdot \left\{ (1 - 1.1r_S) + \frac{r_S^2}{5.4} \right\} \left\{ 1 - \frac{V_{OC}+0.7}{V_{OC}} \cdot \frac{FF_0}{r_{SH}} \cdot \left[(1 - 1.1r_S) + \frac{r_S^2}{5.4} \right] \right\}$$

valid for $r_S < 0.4$; $r_{SH} > 0.4$ and $v_{OC} > 10$

Source: C. Honsberg, S. Bowden, pveducation.org

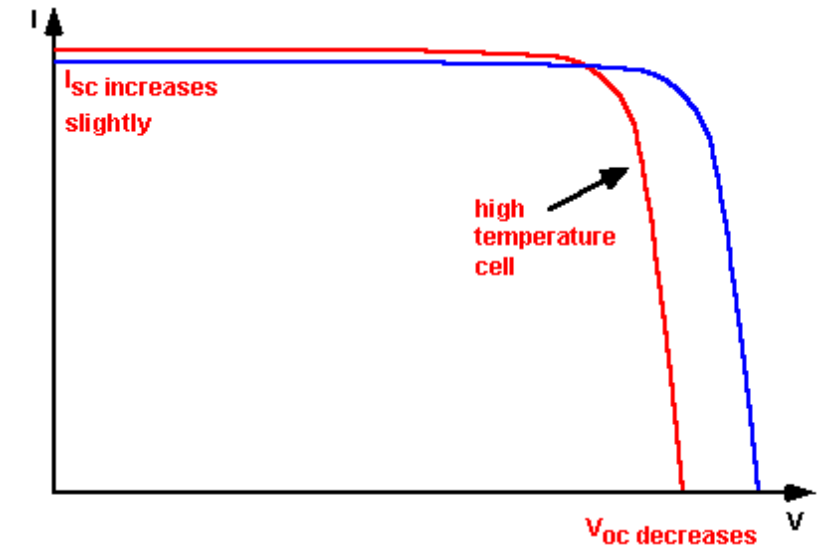
Photovoltaic Cells

Other Influences

- Temperature

- Increased temperature reduces $E_C \rightarrow$ increased n_i
- V_{OC} decreases with temperature; I_{SC} increases slightly
- Influences on Si cells: $dV_{OC}/dT \approx -2.2 \text{ mV/K}$ for Si
 I_{SC} : 0.06 % per K; FF : 1.5 % per K; P_{MP} : 0.4-0.5 % per K

$$V_{OC} = \frac{nkT}{q} \ln \left(\frac{I_L}{I_0} + 1 \right)$$



- Ideality factor

- Usually assumed to be 1
- Depends on recombination mechanism

- Light Intensity

- Changing the number of suns
- 1 sun @ AM1.5: 1000 W/m²
- At low light Intensities: large R_{SH} !!!

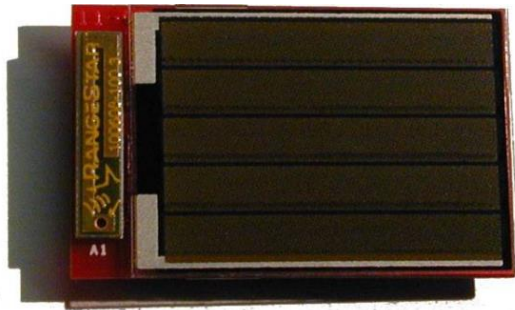
Recombination Type	n	Description
SRH, band to band (low level injection)	1	Recombination limited by minority carrier
SRH, band to band (high level injection)	2	Recombination limited by both carrier types
Auger	2/3	Two majority and one minority carriers required for recombination
Depletion region (junction)	2	two carriers limit recombination

Source: C. Honsberg, S. Bowden, pveducation.org

Photovoltaic Cells

Commercial Solar Cells Types

- Most common are mono- and polycrystalline Si solar cells
- Thin film cells can be produced on different substrates, also flexible substrates are possible (e.g. for wearable solar cells)



Si-thin film
cell on a
glass carrier

© Panasonic

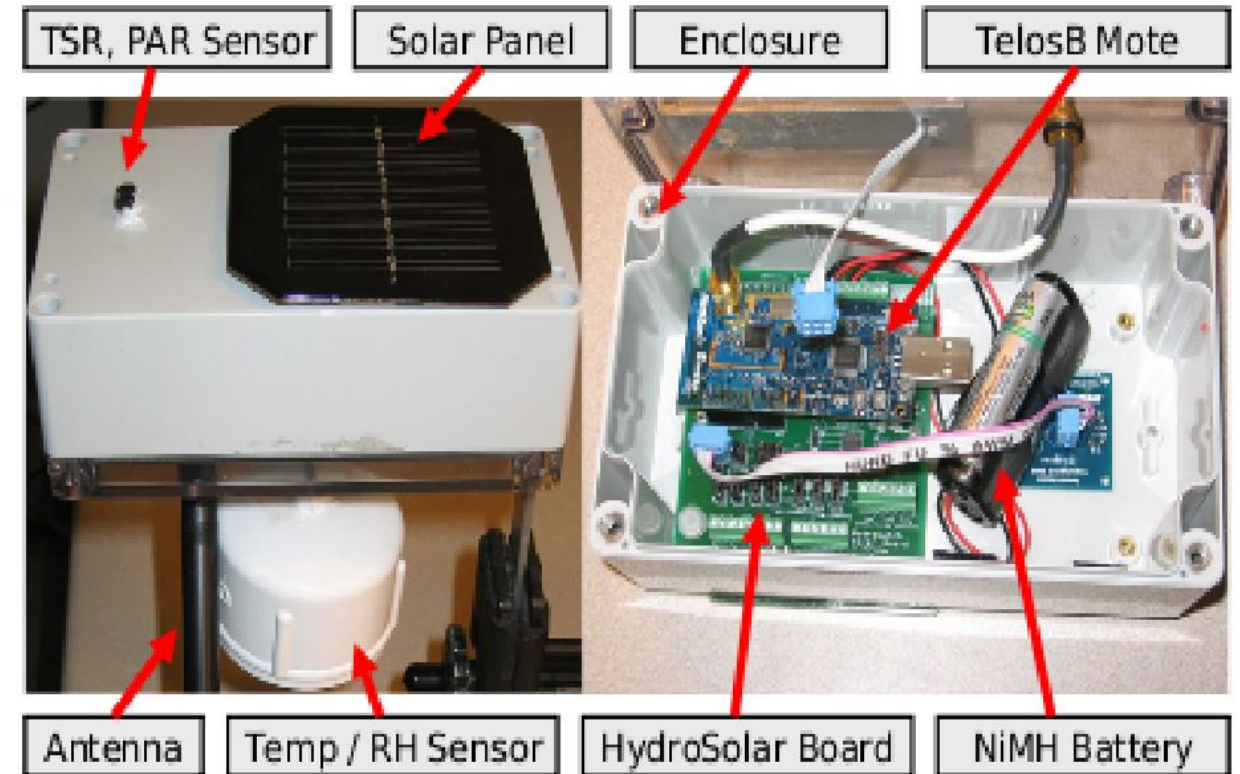
	Cell type	Cell efficiency	Strengths
Gen1: mature technology	Monocrystalline Silicon	25.6 %	Earth abundant, common
	Polycrystalline Silicon	21.3 %	Earth abundant, common
	CIGS (CuInGaSe ₂) (thin films cells)	21.7 %	Flexible Substrates
	CdTe cells (thin films cells)	21.5 %	Flexible Substrates
Gen2: emerging technol.	Dye-sensitized TiO ₂	11.9 %	Tunable colors
	Thin-film Silicon	11.4 %	Flexible modules
	Organic cells	11.5 %	Flexible, semitransparent
	GaAs cells	28.8 %	High efficiency, flexible modules
Gen3	Perovskite	21.0 %	Solution processing

Source: A. Polman et al. *Science* (New York, N.Y.) **2016**, 352, aad4424

Solar-based Energy Harvesting

HydroWatch Weather Node

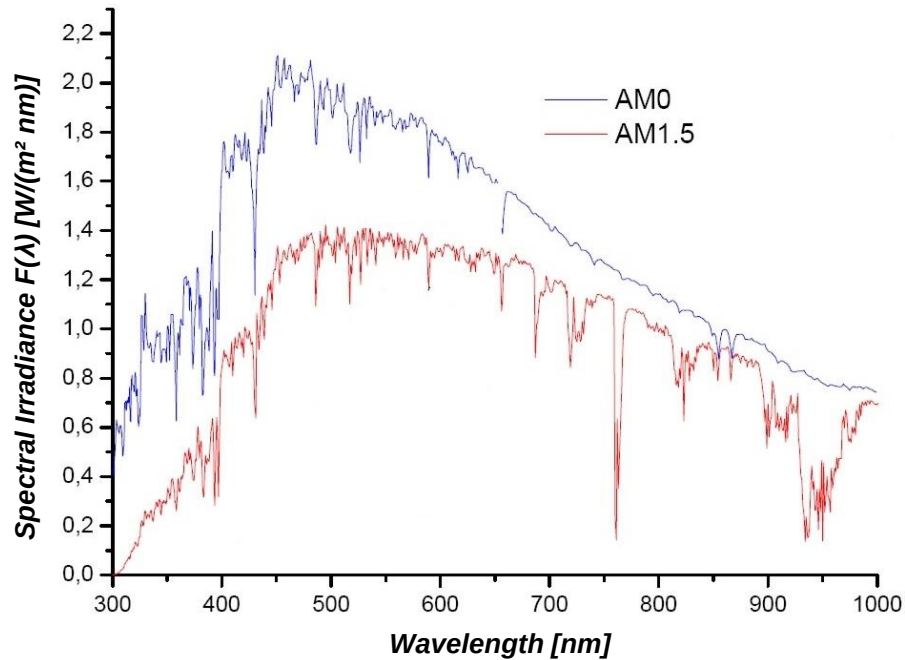
- NiMH batteries enable trickle charging, making the input regulator (50 % efficiency) optional if the panel voltage matches the charging voltage
- System benefits a lot from removing the input regulator
- BUT: high self discharge of NiMH... (30 %) and low faraday efficiency of 66 %
- 30 min of sunlight -> 120 mWh/day (79,2 mWh/day @66 % efficiency)
→ enough for the application!



S. Sudevalayam, P. Kulkarni, *IEEE Commun. Surv. Tutorials* **2011**, 13, 443–461

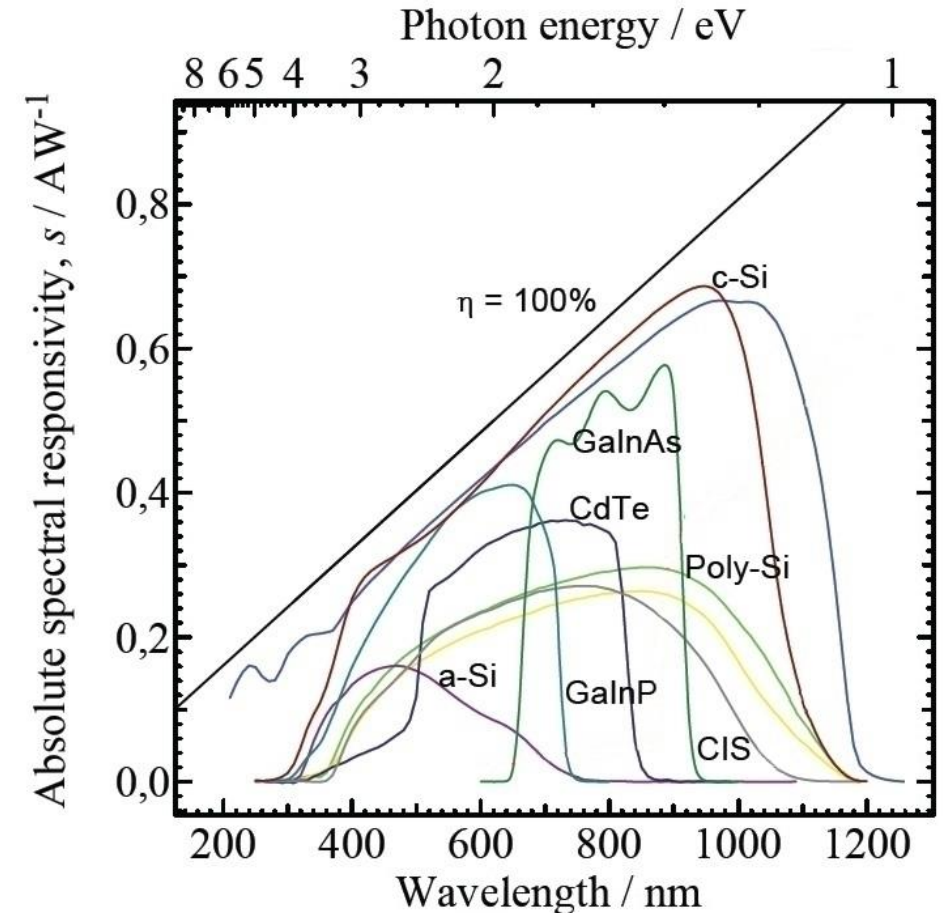
Solar-based Energy Harvesting

a-Si Thin Film Cell



- Calculating P_{max} as function of irradiation and PV technology

$$P_{max} = \int F(\lambda) \cdot QE(\lambda) d\lambda \text{ in W/m}^2$$

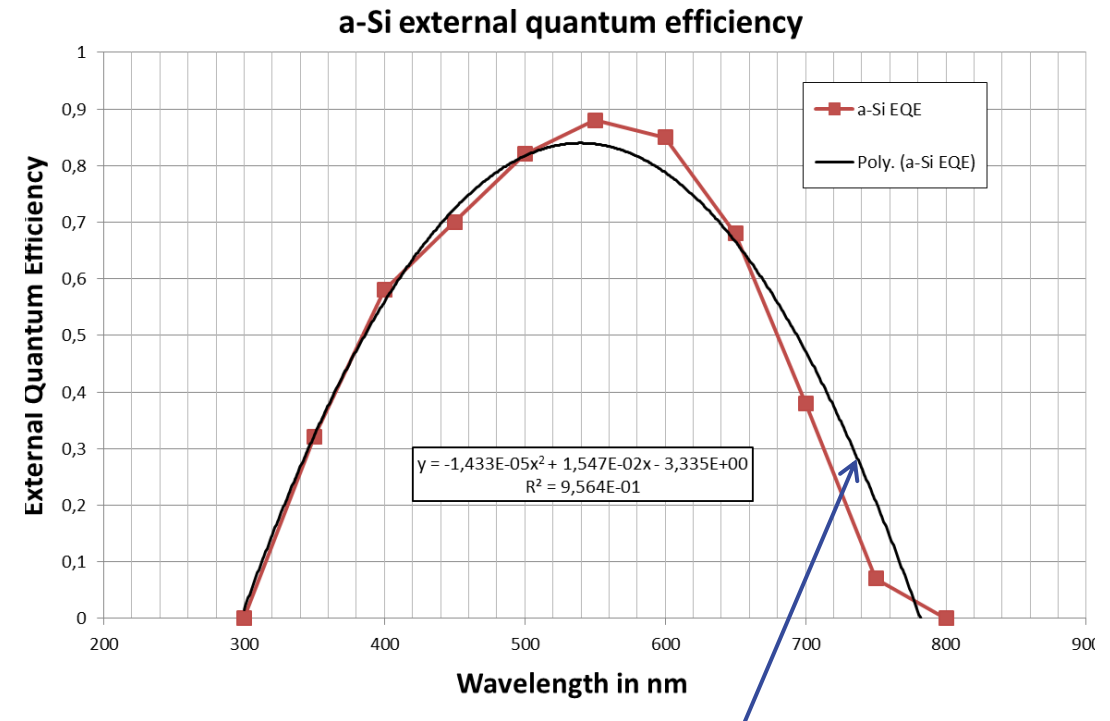


Solar-based Energy Harvesting

a-Si Thin Film Cell

- Measured external quantum efficiency eQE shows absorption between 300 and 800 nm
- A polynomial fit is used to approximate the measured values
- Spectral power density can be calculated as product of eQE and spectral irradiance $F(\lambda)$ (AM1.5G)
- P_{max} per cell area („surface power density“) by integration over the absorption range.

$$P_{max} = \int_{280nm}^{800nm} F(\lambda) \cdot QE(\lambda) d\lambda$$



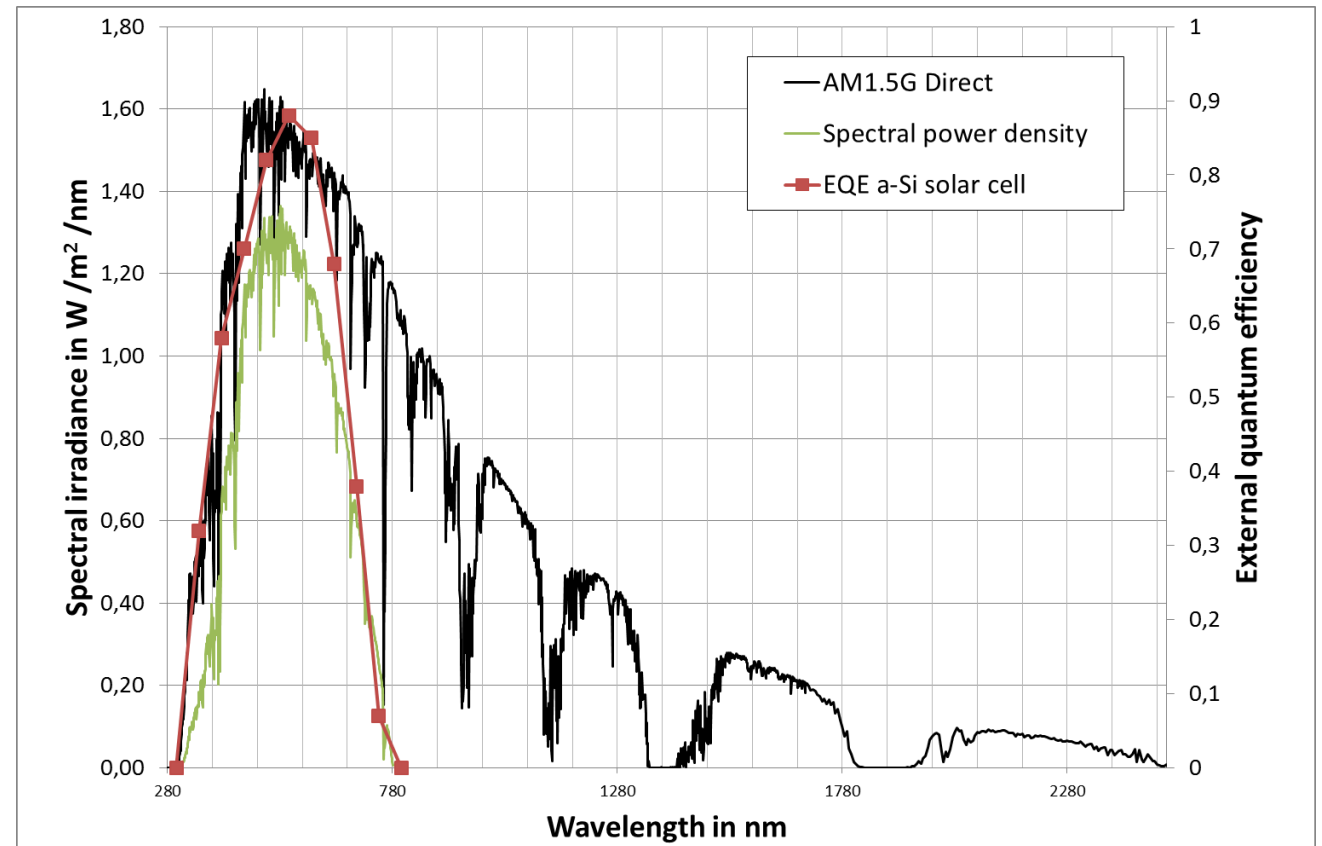
$$f_{it}(\lambda) = -1,433 \cdot 10^{-6} \lambda^2 + 0,0154 \lambda - 3,335$$

$$\int f_{it}(\lambda) d\lambda = 4,7767 \cdot 10^{-6} \lambda^3 + 0,007735 \lambda^2 - 3,335 \lambda$$

Solar-based Energy Harvesting

a-Si Thin Film Cell

- Multiplication of eQE and $F(\lambda)$ is shown as “spectral power density”
- Integration over all absorbed wavelengths (300 - 800 nm) yields the surface power density of the solar cell
- $P_{max} = 188 \text{ W/m}^2$

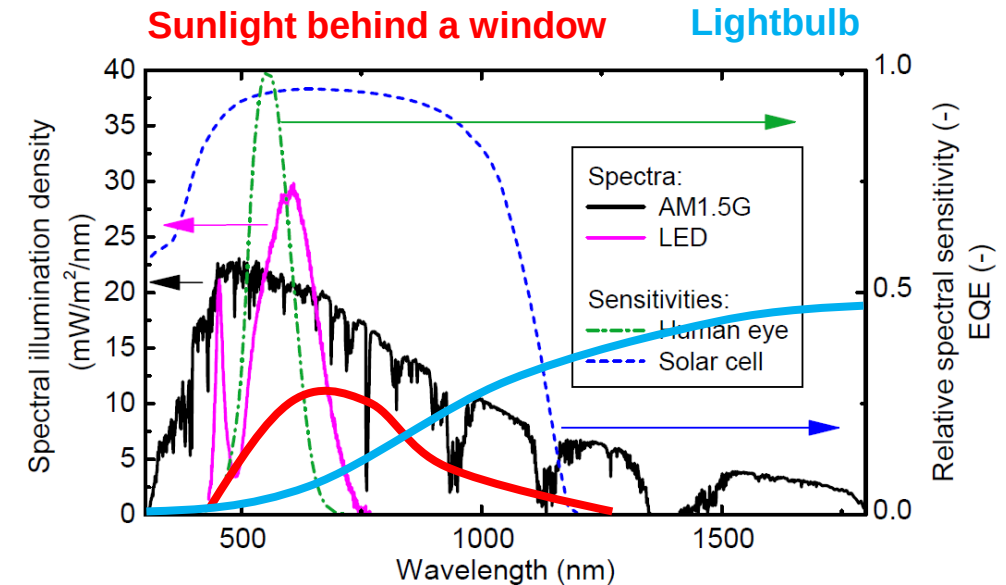


ASTM G173-03 Reference Spectra, www.nrel.gov

Solar-based Energy Harvesting

Indoor use of Solar Cells (IPV – Indoor Photovoltaic)

- Spectral mismatch
 - Indoor light sources have different spectra $\neq$ AM1.5
 - Sunlight passing through windows is reduced in power and the spectrum is shifted
 - Incident light spectrum and spectral response: poor overlap
 - increased number of photons with energies below E_G
 - increased thermalization of excess energy (photons $E > E_G$)
- Available indoor radiation



Sources: K. Rühle, Dissertation, 2015; J. F. Randall, PhD Thesis, EPFL, Lausanne, 2003

Modules 45° tilt, facing south	direct sun	weak clouds	strong clouds	light bulb	fluorescent	power densities with respect to the spectral sensitivity of c-Si-solar cells (solarc, Berlin)
Outdoor, noon	700 - 1200 W/m ²	150 - 500 W/m ²	20 - 50 W/m ²			
indoor, behind glass, noon	500 - 900 W/m ²	50 - 200 W/m ²	5 - 50 W/m ²			
indoor, only artificial light				2 - 10 W/m ²	1 - 3 W/m ²	

Solar-based Energy Harvesting

Specialized Indoor Solar Cells

Datasheet gives current ratings for different illumination and irradiation levels

- Lux: specified for the main sensitivity range of the human eye (~green)
- Irradiance in W/m^2

New ref.		CURRENT / VOLTAGE at maximum power point							
Product ref.	LOW ILLUMINATION		HIGH ILLUMINATION		Dimensions (mms)			Weight (g)	
	200 lux	1000 lux	200 W/m²	1000 W/m²	Length	Height	Thick.*		
3-cells	sous 1.2 V	sous 1.3 V	sous 1.5 V						
03/017/013	4 µA	20 µA	0.5 mA	2 mA / 1.8V	17	13	1.1	0.6	
03/048/016	17 µA	95 µA	2 mA	8 mA / 1.5V	48	16	1.1	2	
03/048/032	40 µA	200 µA	5 mA	18 mA / 1.5V	48	32	1.1	4	

Source: solems.com

Solar-based Energy Harvesting

Considerations for Selecting Solar Cells

- application site (indoor or outdoor ?) ➡ relevant spectra (AM 1.5, artificial light,...)
- irradiation at the application site ➡ direct irradiation, diffuse light, day-night, summer-winter,...
- other boundary conditions ➡ temperature, vibration, pollution, humidity,...